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Grade	**0.00** out of 150.00 (**0**%)

Question 1

Not answered

Marked out of 4.00

Theme- Half-lives of antidepressants
Lead In: Please correctly select the appropriate half-life duration for each of the below medications.

1. Fluoxetine ✖
2. Paroxetine ✖
3. Citalopram ✖
4. Venlafaxine ✖

5 hours
16 hours
30 hours
35 hours
4-5 hours
1-2 hours

Explanation:

1) Fluoxetine has a half-life of around 5-7 days. Its active metabolite, norfluoxetine, has a half-life of 4-16 days, depending on the length of administration 2) Paroxetine- Usually quoted as around 24-30 hours. 3) Citalopram- Usually quoted as around 36 hours. 4) Venlafaxine- Usually quoted as 4-7 hours.

Ref: https://www.nps.org.au/australian-prescriber/articles/switching-and-stopping-antidepressants#t1

The correct answer is: Fluoxetine - 4-5 days; Paroxetine - 30 hours; Citalopram - 35 hours; Venlafaxine - 5 hours

Question 2

Not answered

Marked out of 4.00

Theme- Mode of action

Lead In: Which of the following are the mode of action for each of the below medications?

1. Varenicline ✖
2. Buprenorphine ✖
3. Naltrexone ✖
4. Disulfiram ✖

mu-opioid receptor competitive antagonist.
Aldehyde dehydrogenase competitive antagonist
mu-opioid non-competitive antagonist
Nicotinic acetylcholine receptor partial agonist.
Mu-opioid partial agnoist
d-opioid receptor agonist
Nicotinic acetylcholine receptor antagonist.
Aldehyde dehydrogenase non-competitive antagonist.

Explanation:

Varenicline is an acetylcholine receptor partial agonist. It is an alpha-4 beta-2 partial agonist at nicotinic acetylcholine receptors.

Buprenorphine is a partial agonist at mu-opioid receptors.

Naltrexone is a competitive antagonist of opioid receptors. It is a relatively weak antagonist of kappa- and delta-receptors and a potent mu-receptor antagonist.

Disulfiram is an aldehyde dehydrogenase inhibitor. It irreversibly inhibits aldehyde dehydrogenase.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 837.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 478.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 690, 693.

The correct answer is:

Varenicline - nicotinic acetylcholine receptor partial agonist

Buprenorphine - mu-opioid partial agonist

Naltrexone - mu-opioid competitive antagonist

Disulfiram - aldehyde dehydrogenase non-competitive antagonist

Question 3

Not answered

Marked out of 3.00

Drug-related adverse effects

For the drugs given below, identify the most important adverse effect leading to highly restricted use and requirements for appropriate monitoring

Pimozide	Choose...
Vigabatrin	Choose...
Clozapine	Choose...

Cheese reactions
QT prolongation on ECG
Guillian-Barre syndrome
Haemolytic anaemia
Visual field defects
Blood dyscrasias
Aplastic anaemia
Hypersensivity reactions
Agranulocytosis
Hepatotoxicity

Your answer is incorrect.

Explanation:

Pimozide can increase the QTc interval and potentially cause arrhythmia or sudden death, especially in combination with drugs that raise its levels. Because of its effects on the QTc interval, pimozide is not intended for use unless other options for tic disorders or psychotic disorders have failed. Sertindole is another antipsychotic drug associated with sudden cardiac deaths. Droperidol and Thioridazine are also associated with significant QT prolongation on ECG.

Although a highly effective medication for treatment of infantile spasms and refractory complex seizures, **vigabatrin** has been limited in its use because of reports of bilaterally constricted visual fields.

The prevalence of agranulocytosis during **clozapine** treatment is 0.4%, lower than previously thought and risk of death resulting from this is 0.05%: a rare event. Over 80% of cases of agranulocytosis with clozapine develop within the first 18 weeks of treatment.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 630-631.

Sergott RC. Vigabatrin-associated visual field loss: past, present and future. Expert Review of Ophthalmology. 2014 Jun 1;9(3):145-8.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 236.

The correct answer is: Pimozide → QT prolongation on ECG, Vigabatrin → Visual field defects, Clozapine → Agranulocytosis

Question 4

Not answered

Marked out of 3.00

Chemical class of psychotropic drugs

Match each chemical class below with the most appropriate drug from the given list:

Thienobenzodiazepine	Choose...
Dibenzodiazepine	Choose...
Substituted benzamides	Choose...

Chlorpromazine
Trifluoperazine
Quetiapine
Zuclopenthixol
Amisulpride
Clozapine
Molindone
Risperidone
Loxapine
Haloperidol
Olanzapine

Your answer is incorrect.

Explanation:

Olanzapine is a thienobenzodiazepine with antagonism for multiple receptors. Thienobenzodiazepines are characterised as multi- receptor-targeted- acting- agents. Olanzapine is an atypical antipsychotic agent that is structurally related to clozapine.

Clozapine is a dibenzodiazepine derivative.

Sulpiride and amisulpride are substituted benzamides that are used as antipsychotics.

Dibenzothiazepine - Quetiapine.

Benzisoxazole derivatives include Risperidone.

Thioxanthenes include Flupenthixol and Zuclopenthixol.

Ref: Mendonça Júnior FJ et al. Benzo-and thienobenzo-diazepines: multi-target drugs for CNS disorders. Mini reviews in medicinal chemistry. 2015 Jun 1;15(8):630-47.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 218.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 3.

The correct answer is: Thienobenzodiazepine → Olanzapine, Dibenzodiazepine → Clozapine, Substituted benzamides → Amisulpride

Question 5

Not answered

Marked out of 3.00

Cerebral arteries & lesions

Using the clinical descriptions below, identify the artery most likely to be involved:

A 58-year-old gentleman presents with diplopia, ataxia, nystagmus on looking to the side of the lesion and Horner's syndrome

Cho

A 57-year-old gentleman who is a right handed individual presenting with right hemiparesis, dysphasia and sensory loss involving face & arms on the right side.

Cho

A 67-year-old gentleman with presenting with right-sided weakness especially in the legs. The weakness is milder in the arm, and the face is spared.

Cho

Right anterior cerebral artery
Posterior inferior cerebellar artery
Vertebral artery
Right middle cerebellar artery
Vertebral artery
Right middle cerebral artery
Left anterior cerebral artery
Left middle cerebral artery
Bilateral anterior cerebral artery
Basilar artery
Subclavian artery proximal to ipsilateral vertebral artery.
Posterior cerebral artery.

Your answer is incorrect.

The lateral medullary syndrome or posterior inferior cerebellar artery (PICA) thrombosis, or Wallenberg's syndrome, is a brain-stem infarction presenting as acute vertigo with cerebellar signs. It follows thromboembolism in the PICA or its branches, vertebral artery thromboembolism or dissection. On the side of the lesion, there could be facial numbness (5th nerve), diplopia (6th nerve), nystagmus, ataxia (cerebellar), Horner's syndrome, 9th& 10th nerve lesions. On the contralateral side, there could be spinothalamic sensory loss and mild if any, hemiparesis.

Middle cerebral artery lesions are characterized by hemiplegia, hemianesthesia. If the lesion involves dominant hemisphere, the patients often present with other symptoms such as dysphasia and dyspraxia. If the lesion involves non-dominant hemisphere, then contralateral neglect is highly likely.

Anterior cerebral artery lesions involve contralateral hemiplegia/hemiparesis with weakness involving more of the legs than the arms, and the face will usually be spared.

The correct answer is: A 58-year-old gentleman presents with diplopia, ataxia, nystagmus on looking to the side of the lesion and Horner's syndrome → Posterior inferior cerebellar artery, A 57-year-old gentleman who is a right handed individual presenting with right hemiparesis, dysphasia and sensory loss involving face & arms on the right side. → Left middle cerebral artery, A 67-year-old gentleman with presenting with right-sided weakness especially in the legs. The weakness is milder in the arm, and the face is spared. → Left anterior cerebral artery

Question 6

Not answered

Marked out of 3.00

Tau Protein
Ubiquitin
Amyloid A protein
Beta-amyloid protein
Alpha-synuclein protein
Iron granules
FUS protein

Neuropathology

The subtypes of dementia are associated with abnormal molecules accumulating in the cerebrum. Which of the following molecules are most likely be found in the below types of dementia?

1. Mrs. Gasker is a 72-year-old woman who has attended her GP as her family is increasingly worried about her ability to manage at home due to cognitive problems, although her memory appears to be intact. They have also noted that she has been experiencing tremor and limb stiffness. She has started to have falls in the last few weeks. (Please choose TWO answers)

2. Mrs. Hunter is a 76-year-old woman who has been exhibiting increasingly odd behaviour over the past 18 months. Her family describe her as being "sensitive" and "diligent" but she has started to make lewd jokes and indifferent to the concerns of others. Her husband has noted that she is finding it difficult to look after herself properly. Please choose ONE option.

Explanation:

1) The first case describes a history of dementia with Lewy bodies. The key constituent of Lewy bodies is **alpha-synuclein.** When aggregates of misfolded, overexpressed alpha-synuclein accumulate in neurons (axonal processes and synapses) or surrounding glial cells, Lewy bodies are formed. Lewy bodies are immunoreactive for alpha-synuclein as well as for **ubiquitin** and alpha-beta-crystallin. Neuropathic changes of dementia with Lewy bodies include the presence of Lewy bodies and Lewy neurites in the cerebral cortex as well as in the brainstem. These are often accompanied by some degree of Alzheimer disease neuropathologic changes, often with a moderate beta-amyloid plaque burden but relatively lower tangle burden.

2) The second case describes an example of fronto-temporal dementia. This is associated with **tau protein** deposition, particularly in Pick's disease. Pick disease is characterised by the presence of Pick bodies: rounded, homogeneous neuronal cytoplasmic inclusions, faintly visible on H&E staining. Immunohistochemical staining shows that these inclusions contain 3Rtau. Pick bodies are also strongly argyrophilic with the Bielschowsky technique.

Ref: Sanford AM. Lewy body dementia. Clinics in geriatric medicine. 2018 Nov 1;34(4):603-15.

Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. pp. 6-7, 192-193, 197.

The correct answer is:

1) Mrs. Gasker - Alpha-synuclein protein, Ubiquitin

2) Mrs. Hunter - Tau protein

Question 7

Not answered

Marked out of 3.00

Alpha synuclein a
Alpha synuclein b
Tau
Ubiquitin
Beta amyloid
C9orf72

Neuropathology

Lead In: Which of the intracellular molecular compounds from the given list are most likely to be associated with the conditions below?

1. Mrs Jenny Suplee is a 73-year-old woman who has been brought to the GP by her husband with concerns that she is frequently forgetting words and recent events. She has gotten lost near her local high street and has misplaced items around the house. She is aware that she is struggling with her memory. (Choose ONE) ✖

2. Mr Kevin Broadwater is a 70-year-old man who has been referred to his local psychiatry clinic after developing behavioural changes. His family have noticed that he is being very blunt and rude with others, which is out of character for him. He has been banned from his local pub after bothering a group of young women who were drinking there. (Choose TWO) ✖ ✖

Explanation:

1) This description fits well with Alzheimer's disease, which is associated with intracellular accumulation of hyperphosphorylated tau protein, forming neurofibrillary tangles, and amyloid plaques outside and around neurones.

2) This is most likely to represent frontotemporal dementia, which is associated with intracellular accumulations of tau and ubiquitin.

The correct answer is:

1. Mrs Jenny Suplee - Tau

2. Mr Kevin Broadwater - Tau, Ubiquitin

Click here to see a short video:

Question 8

Not answered

Marked out of 3.00

Theme- Hormones

Lead In: Which of the following hormones are released from each of the below areas?

1. Paraventricular nucleus of hypothalamus ✖

2. Peripheral fat cells ✖

3. Lateral hypothalamus ✖

Leptin
Insulin
Ghrelin
Cholecystokinin
Thyriod stimulating hormone
Follicle stimulating hormone
Corticotropin releasing hormone
Testosterone
Orexin

Explanation:

1) Corticotropin-releasing factor (CRF) neurons in the hypothalamic paraventricular nucleus (PVN) initiate hypothalamic-pituitary-adrenal axis activity through the release of CRF into the portal system as part of a coordinated neuroendocrine, autonomic, and behavioral response to stress. The recent discovery of neurons expressing CRF receptor type 1 (CRFR1), the primary receptor for CRF, adjacent to CRF neurons within the PVN, suggests that CRF also signals within the hypothalamus to coordinate aspects of the stress response. (Ref: Excerpt from J Neurosci. 2018 Feb 21;38(8):1874-1890)

2) White adipose tissue has been identified as secreting leptin, various cytokines, adiposin, acylation-stimulating protein, angiotension, adiponectin, resistin and some others.

3) The lateral hypothalamus regulates energy homeostasis, feeding behavior, arousal, and brain reward mechanisms. The hypocretins (orexins) and melanin-concentrating hormone (MCH) are produced in two distinct populations of neurons in the lateral hypothalamus.

The correct answer is:

1. Paraventricular nucleus of hypothalamus - Corticotropin releasing hormone

2. Peripheral fat cells - Leptin

3. Lateral hypothalamus - Orexin

Question 9

Not answered

Marked out of 3.00

Theme- Receptors

Lead In: Which of the receptors in the given list are primarily acted upon by the drugs below?

1. Buspirone ✖
2. Agomelatine ✖
3. Mirtazapine ✖

5-HT1A
5-HT1B
5-HT2B
5-HT3A
5-HT4B
5-HT5A
5-HT2A

Explanation:

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5-HT1A partial agonist.

Agomelatine has agonist actions at melatonin 1 (MT1) and melatonin 2 (MT2) receptors and antagonist actions at 5HT-2C and 5HT-2B receptors.

Mirtazapine is a multifunctional drug with five principal mechanisms of action: 5HT-2A, 5HT-2C, 5HT-3, alpha-2 adrenergic, and H1 histamine antagonism.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 672-673.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 306-309.

The correct answer is:

1. Buspirone - 5HT1A
2. Agomelatine - 5HT2B
3. Mirtazapine - 5HT2A

Question 10

Not answered

Marked out of 3.00

1p; 4p; 8q; 17q;

14p; 19q; 22q

Theme- Chromosomal disorders

Lead In: Abnormalities in which of the above chromosomal areas would be most likely to cause each of the following clinical presentations?

1. Mrs Frances Lewis is a 45-year-old woman who has been struggling with feeling vague in her thinking for some months. She has now developed odd writhing movements in her limbs. Her father had similar difficulties although they started when he was older. ✖

2. Mr George Park is a 72-year-old man who has presented to his GP due to his memory being intermittently poor. On questioning he informs the GP that he has also been falling over occasionally and has been seeing odd things at times. ✖

3. Mr James Quinn is a 65-year-old man whose families have become concerned, as he has been progressively acting more out of character for the last several months. He has been irritable, ignoring social norms and markedly impulsive. ✖

Explanation:

1) The diagnosis is Huntington's disease that is associated with a CAG repeat mutation on the short arm of chromosome 4.

2) Dementia with Lewy Bodies is associated with the ApoE, located on chromosome 19q.

3) Frontotemporal dementia is associated with the progranulin and MAPT genes, both of which are located on the long arm of chromosome 17.

Ref: Orme T, Guerreiro R, Bras J. The Genetics of Dementia with Lewy Bodies: Current Understanding and Future Directions. Current Neurology and Neuroscience Reports; 2018;18(10):67

The correct answer is:

1. Mrs Frances Lewis - 4p

2. Mr George Park - 19q

3. Mr James Quinn - 17q

Question 11

Not answered

Marked out of 4.00

Mahler's Stages of Development
For the situations given below, identify the Mahler's stage of development.

1. The infant moves autonomously and explores the outer world.

2. The infant feels as though he is completely in his own world.

3. The infant begins to gain awareness that s/he is a different entity from the mother.

Normal austic phase
Normal symbiotic phase
Differentiation
hatching
Practicing
Rapprochement- Initiation
Rapprochement- Crisis
Rapprochement- Solution

Explanation:

1) Practicing occurs between 10 and 18 months. During this phase, the infant is able to crawl and eventually walk, exploring the outer world.

2) Normal autistic phase – during this stage (birth-2 months) the infant is uninterested in external stimuli, and is detached from the outer world.

3) Differentiation/hatching (5-10 months) is the stage where the child becomes aware that s/he is of a different entity from the mother.

Stage	Age	Description: Mahler's separation-individuation
Normal autism	Birth-2 months	Periods of sleep outweigh periods of arousal Reminiscent of intrauterine life
Symbiosis	2-5 months	Developing perceptual abilities enables infant to distinguish inner from outer world Mother-infant perceived as a single fused entity
Differentiation/ Hatching	5-10 months	Progressive neurological development and increased alertness Infant's attention drawn away from self to the outer world Physical and psychological distinctiveness from mother gradually appreciated
Practicing	10-18 months	Ability to move autonomously increases child's exploration of outer world
Rapprochement	18-24 months	Child slowly realises helplessness and dependence Need for independence alternates with need for closeness Child moves away from mother and comes back for reassurance
Object constancy	2-5 years	Child gradually comprehends and is reassured by permanence of mother and other important people, even when not in their presence

Adapted from Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition, 2022. pp. 892-893.

SPMM COURSE

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 892-893.

The correct answer is:

1. Infant moves autonomously - Practicing
2. Infant is in own world - Normal autistic phase
3. Infant gains awareness of being a different entity - Differentiation, Hatching

Question 12

Not answered

Marked out of 3.00

Tust vs mistrust
Autonomy vs. shame
Initiative vs. guilty
Industry vs. inferiority
Ego identity vs. role confusion
Intimacy vs. isolation
Generativity vs. stagnation
Ego integrity vs. despair

Theme- Erikson's stages of development
Lead In: Which of the following from Erikson's stages of development apply to each of the below scenarios?

1. Emily Anderson is a 7-year-old girl who was pleased at being praised for working hard at school. ✖

2. Mrs Joan Bakewell is a 70-year-old woman who feels content with her life, despite failing health, past difficulties and an increased level of physical disability. ✖

3. Mr. Elon Musk, aged 47, gives a speech at a graduation ceremony and refers to "making his mark" on the world through creating or nurturing things that will outlast him. ✖

Explanation:

Erik Erikson created an original and highly influential theory of psychological development and crisis occurring in periods that extended across the entire life cycle. Stage 4 takes place from about 5 years to about 13 years. Its focus is **industry versus inferiority**. The child develops industry by learning new skills and takes pride in the things made. A child unprepared for this stage of psychosocial development, either through an insufficient resolution of previous stages or by current interference, may develop a sense of inferiority and inadequacy. Stage 7 takes place from about 40 years to about 60 years. Its focus is **generativity versus stagnation**. Generativity encompasses a protective concern for generations and social institutions, as well as productivity and creativity. The last and 8th stage – **integrity versus despair** – takes place from about 60 years until death. Erikson defined integrity as "the acceptance of one's one and only life cycle and of the persons who have become significant to it as something that had to be and that, by necessity, permitted of no substitutions."

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1071-1076.

The correct answer is:

1. Emily Anderson - Industry vs. inferiority

2. Mrs Joan Bakewell - Ego integrity vs. despair
3. Mr Elon Musk - Generativity vs. stagnation

Question 13

Not answered

Marked out of 3.00

Theme- Stages of child development
Lead In: At what age of normal development would the following milestones be reached?

1. Cooing ✖

2. Speaking 2 word phrases ✖

3. Able to consistently use 3-4 word sentences and build a tower of 6 blocks ✖

3 months
6-9months
9-12 months
12-18 months
18-24 months
2.5-3 years
3.5-4 years

Explanation:

Important stages in verbal development in children are:
Cooing - Around 3-5 months
Babbling - Around 6-9 months
First word - Around 1 year
2-3 word phrases and vocabulary of around 50 words - 18-24 months
3-4 word sentences - 2.5-3 years

The correct answer is:

1. Cooing - 3 months

2. Speaking 2 word phrases - 18-24 months

3. Using 3-4 word sentences - 2.5-3 years

Question 14

Not answered

Marked out of 2.00

Theme- Psychological assessment

Lead In: Which of the above tools would you use to assess the following functions in an older person?

1. Intelligence ✖

2. Selective attention ✖

Hayling and Brixton tests.
Rey-Osterrieth complex figure
WAIS-IV
Bender-Gestalt Test
Stroop test
Digit span
Four boxes test
Dot cancelletion test

Explanation:

1)The current version of Wechsler Adult Intelligence Scale (WAIS-IV) yields an overall IQ score, plus four index scores and scores on component subtests. It has been validated for use for adults up to the age of 90 years.

2)The Stroop effect is the delay and disruption in naming the colours of words printed in non-matching coloured ink, as when the word 'red' is printed in blue ink, the word 'blue' in green ink, and so on. To perform the task it is necessary to ignore the meanings of the printed words and to respond only to the colours in which they are printed, but with experienced adult readers automatic processing tends to occur, causing interference in the processing of information and a significant increase in the time required to name the colours, compared with the time taken to name the colours of meaningless strings of letters. The Stroop test assesses selective attention, processing speed and executive functioning. It is suitable for use in older people.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is:

1. Intelligence - WAIS-IV

2. Selective attention - Stroop test

Question 15

Not answered

Marked out of 3.00

Hyperglycaemia
Hypercalcaemia
Hypocalcaemia
Hypomagnesaemia
Hyponatraemia
Uraemia

Theme- Electrolyte abnormalities
Lead In: Which of the following electrolyte imbalances are most likely to be the cause of the below clinical presentations?

1. Mr. Harry Dresden is a 62-year-old man who has recently been diagnosed with a depressive episode and has been started on Fluoxetine. Since starting it he has been experiencing nausea, vomiting, muscle cramps and marked lethargy. ✖

2. Mr. Edward Rankin is a 50-year-old man with a diagnosis of bipolar affective disorder who has been stable on Lithium for the past 20 years. He has recently been brought to the GP with limb and joint pains, lethargy, problems with his memory and drinking and urinating large amounts of fluid. He insists that he feels unwell physically but fine in his mood. ✖

3. Mr. Trevor Innes is a 48-year-old man with a long history of alcohol dependence. He had informed his family that he had been troubled by back pain over the last few weeks, but has been presenting as lethargic, nauseous and confused over the past few days. ✖

Explanation:

1) This is most likely to represent hyponatraemia which is a relatively common side-effect of most antidepressant varieties, particularly in older populations.

2) This is most likely to be hypercalcaemia, which is more prevalent in individuals taking lithium for a long duration. This electrolyte change results from secondary hyperparathyroidism (i.e. due to the effect of Lithium on the parathyroid glands) although Lithium induced hypercalcaemia can also occur with normal parathyroid levels. Hypercalcaemia is characterised by polydipsia, polyuria, bone pain, abdominal pain, weakness, confusion, depression, renal calculi and arrhythmias.

3) This is most likely to represent uraemia. The patient has likely developed hepatorenal syndrome and has renal failure secondary to cirrhosis or marked liver dysfunction due to long-standing alcohol misuse.

The correct answer is:

1. Mr. Harry Dresden - Hyponatraemia
2. Mr. Edward Rankin - Hypercalcaemia
3. Mr. Trevor Innes - Uraemia

Question 16

Not answered

Marked out of 3.00

Theme- Rating Scales

Lead In: Which of the following rating scales provide the information in the descriptions below?

YBOCS
YMRS
BDI
HoNoS
GAF
AUDIT
ADOS
BPRS

1. Which one measures severity of psychiatric symptomatology and is used in schizophrenia treatment studies?

2. Which one measures overall psychosocial functioning? ✖

3. Which one assesses autism spectrum disorders? ✖

Explanation:

The Brief Psychiatric Rating Scale (BPRS) was developed in the late 1960s as a short clinician-rated scale (based on observation) for measuring the severity of psychiatric symptomatology. It consists of 18 or 24 items, each rated on a 7-8 point scale. It covers a broad range of areas, including thought disturbance, emotional withdrawal and retardation, anxiety and depression, and hostility and suspiciousness. It has been used extensively for decades as an outcome measure in treatment studies of schizophrenia. Reliability of the BPRS is good to excellent when raters are experienced, but this is difficult to achieve without substantial training.

The Global Assessment of Functioning Scale (GAF) is a rating scale measuring overall psychosocial functioning. The maximum score on the GAF is 100.

The Autism Diagnostic Observation Schedule (ADOS) is a semi-structured and lengthy diagnostic interview given by specially trained clinicians that uses standardised data to aid in diagnosis.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 34-38.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 96-98.

The correct answer is:

1. Severity of psychiatric symptomatology - BPRS
2. Overall psychosocial functioning - GAF
3. Autism spectrum disorders - ADOS

Question 17

Not answered

Marked out of 3.00

4 days
1 week
2 weeks
1 month
3 months
6 months

Theme- ICD-11 duration criteria

Lead In: Which of the following durations are necessary for diagnosis of each of the below clinical disorders in ICD-11?

1. Miss Alison Steele is a 23-year-old woman who has been brought to her local A&E by her friends as she has been acting very unusually for her. She has been barely sleeping, constantly going out and bringing home strangers and has talked about making an album with Beyonce. She insists she is "fantastic" and gets extremely irritable with anyone who suggests that she is acting out of character. ✖

2. Mr. Joffra Archer is a 22-year-old man who has presented to his GP with concerns that his parents are trying to interfere with his university studies. He can hear them whispering to him, even though they live 200 miles away. He firmly believes that they are coming into his room and moving his belongings around, even when they insist that they have not been nearby and try to show him evidence for this. ✖

3. Mrs. Jane Plunkett is a 29-year-old woman who has recently and unexpectedly lost her job. She has been tearful and anxious since, particularly around whether she will get another job. She is hopeful about the future and does not feel that she is depressed although she is finding things more difficult. ✖

Explanation:

1. In ICD-11, a manic episode is an extreme mood state lasting at least **one week** unless shortened by a treatment intervention.

2. In ICD-11, symptoms must have persisted for at least **one month** in order for a diagnosis of schizophrenia to be assigned.

3. In ICD-11, adjustment disorder is a maladaptive reaction of an identifiable psychosocial stressor or multiple stressors that usually emerges within **one month** of the stressor. The symptoms typically resolve within 6 months unless the stressor persists for a longer duration.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is:

1. Miss Alison Steele - 1 week
2. Mr. Joffra Archer - 1 month
3. Mrs. Jane Plunkett - 1 month

Question 18

Not answered

Marked out of 1.00

Which personality disorder is the most likely in a 23-year-old man who is sensitive to rebuffs and avoids close contact as a result?

Select one:

- Paranoid personality disorder
- Antisocial personality disorder
- Narcissistic personality disorder
- Schizoid personality disorder
- Schizotypal personality disorder

Your answer is incorrect.

Explanation:

Clinical features of patients with paranoid personality disorder include being suspicious that others are exploiting, harming, or deceiving them; reading hidden demeaning or threatening meanings into benign remarks/events; holding long-standing grudges; believing that others are not loyal or trustworthy; being reluctant to confide in others; feeling that others are attacking their character or reputation; having recurrent suspicions of infidelity.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 562.

The correct answer is: Paranoid personality disorder

Question 19

Not answered

Marked out of 1.00

A 28-year-old man has complained to local council numerous times about his neighbours throwing rubbish into his garden. Being vengeful, he emptied his weekly rubbish collection into their backyard before going in person to council office to make a complaint. He has always had trouble with different neighbours and has changed 9 houses in last 3 years.

Select one:

- Borderline personality disorder
- Schizotypal personality disorder
- Paranoid personality disorder
- Schizoid personality disorder
- Obsessive-compulsive personality disorder

Your answer is incorrect.

Explanation:

Clinical features of patients with paranoid personality disorder include being suspicious that others are exploiting, harming, or deceiving them; reading hidden demeaning or threatening meanings into benign remarks/events; holding long-standing grudges; believing that others are not loyal or trustworthy; being reluctant to confide in others; feeling that others are attacking their character or reputation; having recurrent suspicions of infidelity.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 562.

The correct answer is: Paranoid personality disorder

Question 20

Not answered

Marked out of 1.00

An 8 year old child presents with sudden movements and vocalisations, as well repetitive rituals. What is the most likely diagnosis?

Select one:

- Tourette syndrome
- Autism spectrum disorder (ASD)
- Obsessive-compulsive disorder (OCD)
- Tourette syndrome and OCD
- ASD and OCD

Your answer is incorrect.

Explanation:

Tic disorders are characterised by brief, sudden, repetitive movements (motor tics) or utterances (phonic or vocal tics) that are temporarily suppressible and are usually preceded by a strong urge to perform the tic. The most common cause of childhood-onset tics is Tourette syndrome (chronic motor tic(s) and vocal tic(s)). The most common comorbidities in Tourette syndrome include ADHD, obsessive-compulsive disorder or behaviour (OCD/B) and autism spectrum disorder. Clinical features of OCD are obsessions (intrusive, repetitive, and distressing thoughts or images) and compulsions (repetitive, stereotyped, unnecessary behaviours).

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

Eapen V et al. Comorbidities, social impact, and quality of life in Tourette syndrome. Frontiers in psychiatry. 2016 Jun 6;7:97.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 690.

The correct answer is: Tourette syndrome and OCD

Question 21

Not answered

Marked out of 1.00

In ICD-11, psychotic symptoms may be present in which severity/severities of depression?

Select one:

- Mild, moderate, and severe
- Moderate and severe
- Severe and profound
- Severe only
- Profound only

Your answer is incorrect.

Explanation:

In ICD-11, depressive episodes in depressive or bipolar disorders may be described in detail by using qualifiers indicating the presence of specific symptoms: the melancholic features qualifier, the anxiety symptoms qualifier, the panic attacks qualifier, and the seasonal pattern qualifier. Additionally, depressive episodes can be described according to severity (mild, moderate, or severe) and remission status (in partial or in full remission). For **moderate and severe** depressive episodes, the presence of psychotic symptoms may also be indicated.

Ref: Gaebel W et al. Changes from ICD-10 to ICD-11 and future directions in psychiatric classification. Dialogues in clinical neuroscience. 2020 Mar;22(1):7.

The correct answer is: Moderate and severe

Question 22

Not answered

Marked out of 1.00

Which of the following is a type of narcissistic defences?

Select one:

- Acting out
- Repression
- Projection
- Suppression
- Splitting

Your answer is incorrect.

Explanation:

Persons with narcissistic personality disorder have a grandiose sense of self-importance; they consider themselves special and expect special treatment. They handle criticism poorly and may become enraged when someone dares criticize them, or they may appear utterly indifferent to criticism. They are commonly interpersonally exploitative, and their self-esteem is fragile. They use primitive defence mechanisms, including **projection** which involves attributing one's own internal unacceptable ideas and impulses to an external target, such as another individual, and reacting accordingly to them.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 565.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 892.

The correct answer is: Projection

Question 23

Not answered

Marked out of 1.00

Select one typical feature of depressive episode with psychotic symptoms

Select one:

- Prominent affective symptoms and mood congruent delusions
- Systematised delusions
- Persecutory delusions
- Bizarre delusions
- Non prominent affective symptoms and mood congruent delusions

Your answer is incorrect.

Explanation:

In depression with psychotic symptoms, usually there is pervasive depressed mood (no reactivity) and marked psychomotor disturbance (sometimes to the point of depressive stupor/catatonia) accompanying delusions (commonly) and hallucinations (10-20%). Psychotic symptoms may be categorised as either mood-congruent, that is, in harmony with the mood disorder ('I deserve punishment because I am so bad'), or mood-incongruent, not in harmony with the mood disorder. Patients with mood-incongruent psychotic symptoms are more likely to have a primary psychotic disorder, such as schizophrenia or schizoaffective disorder.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 251.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 380.

The correct answer is: Prominent affective symptoms and mood congruent delusions

Question 24

Not answered

Marked out of 1.00

A 27-year-old soldier returns from war field with distressing feelings, flashbacks, nightmares, and hypervigilance. Which of the following diagnosis is likely?

Select one:

- Generalised anxiety disorder
- Schizophrenia
- Panic disorder
- PTSD
- Dissociative disorder

Your answer is incorrect.

Explanation:

Post-traumatic stress disorder (PTSD) is a severe psychological disturbance following a traumatic event, characterised by involuntary re-experiencing of elements of the event, with symptoms of hyperarousal, avoidance, and emotional numbing.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 402.

The correct answer is: PTSD

Question 25

Not answered

Marked out of 1.00

Clock drawing test can be used to detect

Select one:

- Abnormal time perception
- Episodic memory loss
- Poor attention
- Constructional dyspraxia
- Disorientation

Your answer is incorrect.

A number of cognitive, motor and perceptual functions are required simultaneously for successful completion of the clock drawing test. Clock drawing is a useful screening technique that is sensitive to organisation and planning as well as **constructional ability**.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 53.

The correct answer is: Constructional dyspraxia

Question 26

Not answered

Marked out of 1.00

A 42-year-old man presents with motor clumsiness, poor coordination and unsteadiness of gait. He is apathetic and his thinking process is slowed. He has a strong family history of similar presentations. What is the most likely diagnosis?

Select one:

- CJD
- Lewy body dementia
- Huntington disease
- Alzheimer dementia
- Wilson disease

Your answer is incorrect.

Explanation:

Huntington disease (HD) is a **rare neurodegenerative disorder** of the central nervous system with **autosomal dominant** inheritance. It is **characterised by** chorea, dystonia, rigidity, akathisia, swallowing difficulties, myoclonus, gait and balance difficulties, bruxism, cognitive decline, psychiatric disorders, and behavioural changes (most common = lack of initiative). **Diagnosis** is straightforward if a patient with a **known family history** of HD presents with the typical neurological movement disorder (gene testing **not** required if there is laboratory confirmation of the Huntington's disease gene in the family). In a patient **without a family history, gene testing is essential** but does not give a clinical diagnosis and must of combined with clinical history and physical exam.

Ref: Kumar A et al. Therapeutic Advances for Huntington's Disease. Brain Sciences. 2020 Jan;10(1):43.

Bachoud-Lévi AC et al. International Guidelines for the treatment of Huntington's Disease. Frontiers in neurology. 2019;10:710.

The correct answer is: Huntington disease

Question 27

Not answered

Marked out of 1.00

Which of the following is a good opening question when discussing compliance with treatment?

Select one:

- It is very important that you take your tablets
- Are you taking your tablets?
- How many times have you missed your tablets so far?
- Why are you not taking your tablets?
- Is there any difficulty with your tablets?

Your answer is incorrect.

Asking "Is there any difficulty with your tablets?" is less judgmental and least confrontational among all the other given options.

The correct answer is: Is there any difficulty with your tablets?

Question 28

Not answered

Marked out of 1.00

Mr Z is a 29-year-old gentleman with history of memory loss and personality changes. He also presents with wing beating tremors. The biochemical abnormality to be identified in this case would be

Select one:

- Low serum ceruloplasmin
- Low albumin
- Low serum calcium
- Low serum thyroxine
- Low serotonin

Your answer is incorrect.

Explanation:

Wilson disease (WD) is a genetic disorder of copper metabolism that leads to copper accumulation in various organs, primarily the liver and brain. Clinical symptoms are a result of organ dysfunction due to the direct or indirect effects of **copper accumulation. The underlying mechanism** is the failure of copper transport in hepatocytes with subsequent impairment of incorporation into the ceruloplasmin and excretion into the bile and blood. **Low ceruloplasmin** may be an initial test for WD but has low sensitivity as a sole assessment. WD is typically described as a **hepato-neurological disease**, with focus on signs of liver disease and extrapyramidal symptoms (e.g. wing-beating tremor and dystonia).

Ref: Litwin T et al. Psychiatric manifestations in Wilson's disease: possibilities and difficulties for treatment. Therapeutic advances in psychopharmacology. 2018 Jul;8(7):199-211.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 158.

The correct answer is: Low serum ceruloplasmin

Question 29

Not answered

Marked out of 1.00

A patient is suspected to have parietal damage related to a vascular lesion. Which of the following signs may be elicited in neurological examination?

Select one:

- Bitemporal hemianopia
- Homonymous hemianopia
- One sided blindness
- Upper quadrantonopia
- Lower quadrantanopia

Your answer is incorrect.

Lower homonymous quadrantanopias are usually caused by damage to the optic radiation as it passes through the parietal lobes.

The correct answer is: Lower quadrantanopia

Question 30

Not answered

Marked out of 1.00

An anorexic patient repeatedly purges with laxatives. Which of the following is a possible complication?

Select one:

- Hyponatraemia
- Metabolic acidosis
- Hyperkalaemia
- Metabolic alkalosis
- None of the listed options

Your answer is incorrect.

Explanation:

Laxative abuse in anorexia nervosa can result in metabolic acidosis. In laxative induced diarrhoea, a large amount of bicarbonate may be lost in the stool. With normal kidneys, the lost bicarbonate is replaced effectively, and a serious base deficit does not develop. When there is poor renal blood flow due to hypovolaemia/starvation, base deficit and acidosis develop rapidly. Acidosis also results from excessive production of lactic acid when patients have severe diarrhoea. Hypokalaemic/ hypochloraemic metabolic alkalosis can occur from repeated vomiting.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 414.

The correct answer is: Metabolic acidosis

Question 31

Not answered

Marked out of 1.00

Which test would you use to screen the cognitive functioning of an elderly patient at the bedside?

Select one:

- MoCA
- Addenbrooke cognitive examination
- Wisconsin card sorting test
- Mini-COG
- Clock drawing test

Your answer is incorrect.

Explanation:

The Montreal Cognitive Assessment (MoCA) is a 30-item questionnaire used for bedside cognitive testing. It is increasingly used in preference to the MMSE due to its assessment of a broader range of cognitive domains and its greater sensitivity and specificity for mild cognitive impairment. A score of below 26 suggests impairment. It is available in many languages and in electronic form.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 86.

The correct answer is: MoCA

Question 32

Not answered

Marked out of 1.00

Read the strip of ECG shown below and identify the relevant abnormality. Blue lines indicate PR intervals. A 'P wave' is also marked in the figure.



Select one:

- Complete heart block
- Type I third degree heart block
- Mobitz type II
- Mobitz type I
- Normal ECG

Your answer is incorrect.

Mobitz type I AV block: The PR interval shows progressive lengthening until one P wave fails to be conducted and fails to produce a QRS complex. The PR interval resets to normal, and the cycle repeats. This is second-degree atrioventricular nodal block, also known as Wenckebach block.(Ref: Houghton & Gray, Making sense of the ECG. Pg 120-122)

The correct answer is: Mobitz type I

Question 33

Not answered

Marked out of 1.00

'I saw a blue caravan on the road and I immediately realised that the South Pole is going to sink soon'. This is consistent with

Select one:

- Delusional perception
- Delusional memory
- Delusional atmosphere
- Delusional mood
- Autochthonous delusion

Your answer is incorrect.

Explanation:

A delusional perception is a primary delusion which is recalled as having arisen as a result of a perception (e.g. a patient who, on seeing two white cars pull up in front of his house, became convinced that he was therefore about to be wrongly accused of being a paedophile). The percept is a real external object, not a hallucinatory experience.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 107.

The correct answer is: Delusional perception

Question 34

Not answered

Marked out of 1.00

Cameron described the following form of thought disorder:

Select one:

- Omission
- Asyndesis
- Fusion
- Drivelling
- Desultory thinking

Your answer is incorrect.

Norman Cameron's analysis of formal thought disorder in 25 'unmistakably schizophrenia' patients led to the concept of asyndetic thinking that included derailment, loss of the central determining idea, and loose associations. He emphasised the lack of connections between successive thoughts in this phenomenon of 'asyndesis'. For example, 'I got up at eight this morning as well as few birds of different colours on the painting, shrinking all the time to drop few coins. On the floor. All the time.' Cameron also proposed 3 other characteristic formal thought disorders –

- *Metonymy*: imprecise approximate expressions used as substitute words. For example paperskate for a pen.
- *Overinclusion*: In overinclusive thinking ideas that are only remotely related to the concept under consideration become incorporated in the patient's thinking; Conceptual boundaries are lost.
- *Interpenetration*: Irrelevant thoughts penetrate ongoing stream of thoughts.

He also described *thought fragmentation*, similar to Schneider's *derailment* concept (see below)

Carl (*not* Kurt) **Schneider** proposed a different set of 5 elements of formal thought disorder.

- *Substitution*: one thought – often inappropriate, fills the gap between other appropriate, more consistent thoughts.
- *Omission*: A chunk of thought goes missing from stream of conversation, patient being unaware – best analysed when written.
- *Fusion*: various thoughts fuse together, leading to loss of goal direction.
- *Drivelling*: disordered intermixture of constituent parts of one complex thought
- *Derailment* (aka entgleisen); In derailment normally flowing track of thoughts suddenly change. The determining tendency is preserved but is misdirected.
- Schneider also described *desultory thinking,* sometimes considered along with driveling. In desultory thinking, speech is grammatically correct but sudden ideas force their way in from time to time. Each one of these ideas is a simple thought that, if used at the right time would be quite appropriate.

Ref: Jerónimo J et al. Formal Thought Disorders–Historical Roots. Frontiers in psychiatry. 2018 Nov 9;9:572.

The correct answer is: Asyndesis

Question 35

Not answered

Marked out of 1.00

A patient describes that everything around him appears hazy, dull and lifeless. This can be termed as

Select one:

- Anhedonia
- Aphemia
- Depersonalisation
- Derealisation
- Agnosia

Your answer is incorrect.

Explanation:

Derealisation is an unpleasant subjective experience where the patient feels as if the world has become unreal. It is a feeling that one's environment has changed in some strange way that is difficult to describe. Derealisation is a non-specific symptom of a number of disorders.

Depersonalisation is an unpleasant subjective experience where the patient feels as if they have become 'unreal'. It is a feeling that one is not oneself or that something has changed. Depersonalisation is a non-specific symptom occurring in many psychiatric disorders, as well as in 'normal' people.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 108.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 10.

The correct answer is: Derealisation

Question 36

Not answered

Marked out of 1.00

When asked about his favorite sport the patient answers, "I like soccer. The players wear shirts. Shirts are clothes. Clothes are dirty." The speech disturbance noted here is best described as

Select one:

- Loosening of association
- Paragrammatism
- Tangentiality
- Flight of ideas
- Circumstantiality

Your answer is incorrect.

Explanation:

In flight of ideas thoughts follow each other rapidly; there is no general direction of thinking; and the connections between successive thoughts appear to be due to chance factors which, however, can usually be understood. The patient's speech is easily diverted by external stimuli and internal superficial associations. The absence of a determining tendency to thinking allows the associations of the train of thought to be determined by chance relationships, verbal associations of all kinds, clang associations, proverbs, maxims, and cliches. The chance linkage of thoughts in flight of ideas is demonstrated by the fact that one could completely reverse the sequence of the record of a flight of ideas, and the progression of thought would be understood just as well (or just as poorly).

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 44.

The correct answer is: Flight of ideas

Question 37

Not answered

Marked out of 1.00

The nature of conversational speech in jargon dysphasia is

Select one:

- Non-fluent and intelligible
- Fluent and unintelligible
- The rhythm of speech is not retained
- Non-fluent and unintelligible
- Fluent and intelligible

Your answer is incorrect.

Jargon aphasia is a form of fluent aphasia characterised by a copious flow of unintelligible speech. Jargon-aphasic speech may be subdivided into semantic jargon, in which ordinary words are strung together to form unintelligible utterances, and phonemic jargon involving neologisms.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Fluent and unintelligible

Question 38

Not answered

Marked out of 1.00

According to the World Health Organization, how many women have experienced intimate partner violence or non-partner sexual violence in their lifetime?

Select one:

- 1 in 100
- 1 in 10
- 1 in 3
- 1 in 20
- 1 in 5

Your answer is incorrect.

Explanation:

According to the World Health Organisation, almost 1 in 3 women have experienced one or both of these forms of violence (intimate partner violence or non-partner sexual violence) at least once in their lifetime. This recent data confirms that physical and sexual intimate partner violence and sexual violence more broadly remain pervasive in the lives of women and adolescent girls across the globe.

Ref: World Health Organization. Violence against women prevalence estimates, 2018: global, regional and national prevalence estimates for intimate partner violence against women and global and regional prevalence estimates for non-partner sexual violence against women. 2021.

The correct answer is: 1 in 3

Question 39

Not answered

Marked out of 1.00

Which of the following is an example of an ideology that emphasizes the rights of groups over the individual?

Select one:

- Individualism
- Socialism
- Collectivism
- Democracy
- Communism

Your answer is incorrect.

Group rights, also known as collective rights, are rights held by a group rather than by its individual members, the latter of which is termed individualism.

The correct answer is: Collectivism

Question 40

Not answered

Marked out of 1.00

William and Diane Corndale are seeking help for infertility. They have been trying to conceive for 5 years, but have so far been unsuccessful. Which of the following is the most likely consequence of this stressor?

Select one:

- Feelings of guilt
- Negative impact on self-esteem
- Decrease in frequency of sexual intercourse
- Increase in number of disagreements
- All of the listed options

Your answer is incorrect.

Difficulty in conceiving may place significant stress on a couple, and there's the potential for a wide spectrum of psychological struggles to ensue.

The correct answer is: All of the listed options

Question 41

Not answered

Marked out of 1.00

In his 1987 book, Le Suicide, what did sociologist Emile Durkheim conclude?

Select one:

- The risk of suicide is fixed
- Suicide was more common during times of war
- Suicide rates are higher in women than in men
- Suicide could be attributed to a lack of social cohesion
- Suicide was more common in civilians than in soldiers

Your answer is incorrect.

Explanation:

Emile Durkheim was a French sociologist who challenged the assumption that social processes can be understood in terms of the actions of individuals and argued that even suicide, apparently the ultimately individualistic act, cannot be fully explained at the purely individual level. In his book *Suicide*, Durkheim used survey data purporting to show that egoistic suicide, resulting from feelings of self-reproach and sense of failure, arises from a lack of social cohesion; that altruistic suicide, carried out for the benefit of others, arises from a sense of failure to society; and that anomic suicide, arising from a sense that life is pointless, arises from an absence of social norms that he called anomie.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Suicide could be attributed to a lack of social cohesion

Question 42

Not answered

Marked out of 1.00

Mr Loughton has come to see you in clinic, following a referral from the police. You are aware that your colleague will want to know what his future risk of sexual violence will be. Which of the following is the strongest predictive factor for future sexual violence?

Select one:

- History of bipolar affective disorder
- History of past sexual violence
- History of depression
- Having numerous siblings
- History of alcohol misuse

Your answer is incorrect.

Explanation:

History of sexual violence perpetration is the strongest predictor of sexual violence perpetration.

Ref: Steele B et al. Risk and Protective Factors for Men's Sexual Violence Against Women at Higher Education Institutions: A Systematic and Meta-Analytic Review of the Longitudinal Evidence. Trauma, Violence, & Abuse. 2020 Nov 11:1524838020970900.

The correct answer is: History of past sexual violence

Question 43

Not answered

Marked out of 1.00

Mandip is a 33-year-old Sikh male from India. He is an IT contractor and speaks English at work. He regularly goes to the pub after work for drinks with his colleagues. When he goes home he speaks Punjabi with his parents and wears traditional robes. What form of acculturation does Mandip display?

Select one:

- Integration
- Division
- Assimilation
- Separation
- Marginalisation

Your answer is incorrect.

Acculturation refers to the process of cultural change that takes place when an individual or a group comes in to continuous contact with a culturally distinct group. Acculturation can result from immigration and can occur in either direction – hosts can get accultured; this was evident in certain places in times of Colonial rule. Four types are described according to the degree of retention and adoption of the two cultures at ethnocultural group level. Integration refers to both high retention of one's own cultural values and high adoption of the practices of the new culture.

The correct answer is: Integration

Question 44

Not answered

Marked out of 1.00

Empathy skills are most likely to be delayed in which of the following?

Select one:

- A deaf child of deaf parents
- A hearing child of deaf parents
- A deaf child of hearing parents
- A hearing child with one deaf parent
- A hearing child of hearing parents

Your answer is incorrect.

Many deaf children from hearing families have seriously delayed development of empathy and theory of mind (ToM). They often continue to fail false belief and other standard preschool ToM tests well into middle childhood. However, natively signing deaf children of deaf parents develop ToM understanding on the same early timetable as hearing children. This suggests the importance of shared conversation with family members via a mutually accessible language (either signed or spoken) for the development of ToM. In deaf children of hearing parents empathy and ToM are significantly correlated, unlike in hearing children where these two are less dependent.

Ref: Peterson CC. Empathy and theory of mind in deaf and hearing children. Journal of deaf studies and deaf education. 2016 Apr 1;21(2):141-7.

The correct answer is: A deaf child of hearing parents

Question 45

Not answered

Marked out of 1.00

Michael Stills is 32 years old and has developed depression for the first time in in his life. You meet him and take a history. Which of the following is a precipitating factor for his depression?

Select one:

- Working as a teaching assistant
- Growing up in a large city
- Being married for the past 2 years
- Having a parent who has a diagnosis of chronic anxiety disorder
- The recent loss of his baby son from cot death at the age of 4 months

Your answer is incorrect.

Explanation:

Four 'Ps' of case formulation:

Predisposing factors: those that put a person at risk of developing a problem; may include genetics, previous life events, or temperament

Precipitating factors: a specific event or trigger to the onset of the current problem

Perpetuating factors: those that maintain the problem once it has become established

Protective factors: strengths that reduce the severity of problems and promote healthy and adaptive functioning

Ref: Racine NM et al. Systematic review: predisposing, precipitating, perpetuating, and present factors predicting anticipatory distress to painful medical procedures in children. Journal of pediatric psychology. 2016 Mar 1;41(2):159-81.

The correct answer is: The recent loss of his baby son from cot death at the age of 4 months

Question 46

Not answered

Marked out of 1.00

Selina has had a spider phobia since she was a child. It impairs her ability to function, as she avoids any places whereby she might come across a spider: non-urban areas, public places that do not appear spotless. Her greatest fear is having a spider crawl over her face. She agrees to participate in some talking therapy to help her with her phobia. In one session, the therapist gives a very detailed description of a spider crawling across Selina's face. Her task is to withstand the anxiety until it diminishes in intensity. What method is this describing?

Select one:

- Biofeedback
- Implosion
- Stimulus hierarchy
- Operant conditioning
- Counter conditioning

Your answer is incorrect.

Explanation:

By exposing the patient to the phobic object and preventing the usual escape or avoidance, there is extinction of the usual (maladaptive) anxiety response. This may be done *in vivo* (flooding) or in imagination (implosion).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 908.

The correct answer is: Implosion

Question 47

Not answered

Marked out of 1.00

Two medications have the same efficacy but Drug A is marketed as having a 50% failure rate whilst Drug B is marketed as having a 50% success rate. Drug B is prescribed more frequently. What is this an example of?

Select one:

- Hawthorne effect
- Barnum effect
- Heuristics
- Framing effect
- Outcome bias

Your answer is incorrect.

Explanation:

The framing effect is an example of cognitive bias, in which people react to a particular choice in different ways depending on how it is presented; e.g. as a loss or as a gain. It is an effect of the description, labelling, or presentation of a problem or responses to it. In this question, the two frames are merely different ways of describing the same thing.

Heuristics are simple, efficient rules which people often use to form judgments and make decisions. The outcome bias is an error made in evaluating the quality of a decision when the outcome of that decision is already known. The Barnum effect is a common psychological phenomenon whereby individuals will give high accuracy ratings to descriptions of their personality that supposedly are tailored specifically to them but that are, in fact, vague and general enough to apply to a wide range of people. The Hawthorne effect is the alteration of behaviour by the subjects of a study due to their awareness of being observed.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Framing effect

Question 48

Not answered

Marked out of 1.00

'Internal salience' is a concept in psychology relating to cognitive processes. Which of the following is the most accurate description of it?

Select one:

- A belief that a particular object or person is important
- Rating one's own thoughts as being more important than those of others
- Rating one's own beliefs above factual information
- Self-motivated desire for an object or action
- Yearning for reminders of loved ones or objects

Your answer is incorrect.

Motivational salience is a cognitive process and a form of attention that motivates, or propels, an individual's behaviour towards or away from a particular object, perceived event, or outcome. This can be driven by external factors, such as encouragement or instruction from a teacher or similar, or internal, from within an individual due to their pre-existing beliefs.

The correct answer is: Self-motivated desire for an object or action

Question 49

Not answered

Marked out of 1.00

There are various factors, which influence group dynamics and impact upon their effectiveness. Which of the following factors can reduce the cohesiveness of a group?

Select one:

- Members function independently
- Large group size
- Role ambiguity
- Member diversity
- Open to new members

Your answer is incorrect.

Role ambiguity refers to the lack of clear definition of what is expected of an individual, with respect to the goals of their team. The more uncertainty there is about the tasks, goals and levels of performance, the less satisfied workers tend to be. But in the presence of task and social cohesion job satisfaction improves greatly.

Ref: Urien B, Osca A, García-Salmones L. Role ambiguity, group cohesion and job satisfaction. Revista Latinoamericana de Psicología. 2017 May 1;49(2):137-45.

The correct answer is: Role ambiguity

Question 50

Not answered

Marked out of 1.00

Attention results in the conversion of:

Select one:

- Short term memory into long term memory
- Iconic memory to echoic memory
- Working memory into short term memory
- Sensory memory into short term memory
- Retrieval of long term memory

Your answer is incorrect.

Sensory receptors in the brain are responsible for receiving sensory information which they store for mere seconds. The brain filters this multifaceted sensory information from multiple sensory organs. When conscious attention is paid, this certain sensory information is selected and converted into short-term memory which enables us to remember it.

The correct answer is: Sensory memory into short term memory

Question 51

Not answered

Marked out of 1.00

Working memory can be tested at bedside using which of the following?

Select one:

- Cognitive Estimation Test
- Clock-drawing Test
- Digit span
- Wisconsin Card Sorting Test
- Stroop Test

Your answer is incorrect.

Explanation

Digit span tests working memory by asking the participant to recall sequences of numbers of increasing lengths in a forward or backward manner. It requires patients to repeat longer strings of digits increasingly as a way of assessing the ability to process relatively simple information.

Cognitive estimation is used to test problem solving. Wisconsin card sorting test is used to test set shifting. Stroop test is used to test selective attention. Clock drawing test is used to test general cognitive ability in dementia.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 54.

The correct answer is: Digit span

Question 52

Not answered

Marked out of 1.00

Anna is 8 years old. You are conducting some exercises to determine what stage of development she appears to be at, according to Piaget's stages of development. You present her with two glasses, both of the same shape, and both holding the same amount of orange juice. You then obtain two other glasses of different shapes: one is tall and thin, another short and broad. You pour the orange juice from one of the original glasses into the tall glass, and the juice from the other original glass into the shorter glass. You now ask Anna to tell you which glass holds the most juice. What answer indicates she has reached the concrete operational stage of development?

Select one:

- 'The short glass has the most orange juice'
- 'They each have more orange juice than before'
- 'Glasses can be half empty or half full'
- 'Both glasses have the same amount of orange juice'
- 'The tall glass has the most orange juice'

Your answer is incorrect.

Explanation:

According to Piaget, the stage of concrete operations lasts from about 7 to 11 years of age. In this period, children operate and act on the concrete, real, and perceivable world of objects and events. Egocentric thought is replaced by operational thought, which involves dealing with a wide array of information outside the child. Children can now see things from someone else's perspective. Children also understand the concept of **conservation**, which is the ability to recognise that, although the shape of objects may change, the objects still maintain or conserve other characteristics that enable them to be recognised as the same. The ability of children to understand quantity is one of Piaget's most important cognitive developmental theories. Measures of quantity include measures of substance, length, number, liquids, and area.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1008-1009.

The correct answer is: 'Both glasses have the same amount of orange juice'

Question 53

Not answered

Marked out of 1.00

Which of the following of the Kohlberg's stages of moral development are paired correctly?

Select one:

- Pre-conventional - punishment
- Post-conventional - conformity
- Post-conventional – authority
- Pre-conventional – social contract
- Conventional – universal principles

Your answer is incorrect.

Explanation:

Lawrence Kohlberg integrated Jean Piaget's concepts and described multiple stages of moral development within three significant levels of morality. The first level is **pre-conventional** morality, in which **punishment** and obedience are the determining factors. The second level is morality of conventional role-conformity, in which children try to conform to gain approval and maintain good relationships with others and are also guided by rigid codes of 'law and order' (authority orientation). The third and highest level is the morality of self-accepted moral principles (post-conventional), in which children voluntarily comply with rules based on a concept of universal ethical principles and make exceptions to rules in certain circumstances.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 902.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Pre-conventional - punishment

Question 54

Not answered

Marked out of 1.00

The Oedipus complex is associated with which stage of psychosexual development?

Select one:

- Phallic
- Latency
- Oral
- Genital
- Anal

Your answer is incorrect.

Explanation:

The phallic stage begins sometime during year 3 and continues until approximately the end of year 5. Threats of castration and the related anxiety are connected with guilt over masturbation and oedipal wishes. During this phase oedipal involvement and conflict are established and consolidated.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1062.

The correct answer is: Phallic

Question 55

Not answered

Marked out of 1.00

Traditionally, the view has been that boys like to play with cars and girls like to play with dolls. Which of the following is this an example of?

Select one:

- Normalisation
- Socialisation
- Indoctrination
- Stereotyping
- Internalisation

Your answer is incorrect.

Explanation:

Socialisation is the process, beginning in infancy, whereby one acquires the attitudes, values, beliefs, habits, behaviour patterns, and accumulated knowledge of one's society, through child-rearing, education, and modification of one's behaviour to conform with the demands of the society or group to which one belongs.

Indoctrination is the process of inculcating a person with ideas, attitudes, cognitive strategies or professional methodologies. A stereotype is any thought widely adopted about specific types of individuals or certain ways of behaving intended to represent the entire group of those individuals or behaviors as a whole. Normalisation refers to social processes through which ideas and actions come to be seen as "normal" and become taken-for-granted or "natural" in everyday life. Internalisation is an individual's acceptance of a set of norms and values via socialisation.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Socialisation

Question 56

Not answered

Marked out of 1.00

Sharon lives in the same household with her younger brother who is in a same-gender partnership. Her brother and his partner want to adopt a child. Which of the following statements regarding lesbian and gay (LG), and heterosexual adoptive parents is true?

Select one:

- Lesbian mothers, heterosexual mothers, gay fathers, and heterosexual fathers report no differences in parenting competence in adoptive families
- Heterosexual couples are more likely to adopt children than same-gender couples
- Adoptive parents of all types no longer face stigma in our society
- LG adoptive parents face additional stigma based on sexual identity
- Stigmatisation based on sexual identity has no effect on adoptive parents' mental health

Your answer is incorrect.

Explanation:

Adoptive parents of all types often face stigma related to 'non-traditional' family structures. Lesbian and gay (LG) adoptive parents often experience additional stigmatisation based on sexual identity, which in turn may negatively affect parents' mental health. Children and their adoptive sexual minority parents face stigma, overt discrimination, and microaggressions based on parent sexual orientation. Lesbian mothers, heterosexual mothers, and gay fathers all report higher parenting competence than heterosexual fathers in adoptive families. Same-gender couples may be up to seven times more likely to have adoptive children than different-gendered couples.

Ref: Farr RH, Vázquez CP. Stigma Experiences, Mental Health, Perceived Parenting Competence, and Parent–Child Relationships Among Lesbian, Gay, and Heterosexual Adoptive Parents in the United States. Frontiers in psychology. 2020 Mar 30;11:445.

The correct answer is: LG adoptive parents face additional stigma based on sexual identity

Question 57

Not answered

Marked out of 1.00

Which of the following Erikson's stages of development are paired correctly?

Select one:

- 0-1.5 years old – autonomy vs. shame
- 40-65 years old – ego integrity vs. despair
- 12-18 years old – industry vs. inferiority
- 3-5 years old – initiative vs. guilt
- 1.5-3 years old – trust vs. mistrust

Your answer is incorrect.

Explanation:

Erik Erikson created an original and highly influential theory of psychological development and crisis occurring in periods that extended across the entire life cycle. Stage 3 takes place from **about 3 years to about 5 years**. Its focus is **initiative versus guilt**. The child's increasing mastery of locomotor and language skills expands its participation in the outside world and stimulates omnipotent fantasies of broader exploration and conquest. Guilt over the drive for conquest and anxiety over the anticipated punishment are both assuaged in the child through repression of the forbidden wishes and development of a superego to regulate its initiative.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1074.

The correct answer is: 3-5 years old – initiative vs. guilt

Question 58

Not answered

Marked out of 1.00

James Williams is a 2-year-old boy who is walking confidently. When he attempts to run he falls over and becomes very tearful and upset. According to Erikson's Psychosocial model, which of the following crises is he most likely to be experiencing?

Select one:

- Autonomy vs shame
- Autonomy vs inferiority
- Initiative vs guilt
- Industry vs inferiority
- Basic trust vs basic mistrust

Your answer is incorrect.

Explanation:

Erik Erikson created an original and highly influential theory of psychological development and crisis occurring in periods that extended across the entire life cycle. Stage 2 takes place from **about 18 months to about 3 years**. Its focus is **autonomy versus shame and doubt**. Erikson believed that this stage "becomes decisive for the ratio between loving goodwill and hateful self-insistence, between cooperation and willfulness, and between self-expression and compulsive self-restraint or meek compliance". Where that ratio is favourable, the child develops an appropriate sense of autonomy and the capacity to 'have and to hold'; where it is unfavourable, doubt and shame will undermine free will.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1074.

The correct answer is: Autonomy vs shame

Question 59

Not answered

Marked out of 1.00

You are working with a Child and Adolescent Mental Health team, and reading about child psychosexual development. Sigmund Freud proposed a number of concepts concerning psychosexual development. Which of the following is true regarding Freud's theory?

Select one:

- The phallic stage precedes the anal stage
- The Oedipus complex describes a boy's psychosexual competition with his mother for possession of his father
- Between the ages of 6 and puberty, the child's libido is dormant
- The genital and the phallic stages are interchangeable
- In the oral phase the Oedipus complex comes into play

Your answer is incorrect.

Explanation:

The latency stage is the stage of relative instinctual quiescence or inactivity of sexual drive during the period from the resolution of the Oedipus complex until pubescence (from about 5-6 years until about 11-13 years). The oral phase (age 0-1) is followed by the anal (2-3), then phallic (3-5) during which the Oedipus complex becomes activated, then latency (6-puberty), then genital (puberty).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1063.

The correct answer is: Between the ages of 6 and puberty, the child's libido is dormant

Question 60

Not answered

Marked out of 1.00

Ms Coyle tells you that Lauren, her second eldest child, behaved in a puzzling and often upsetting way at times when she had to leave her for. She describes that Lauren would cry very dramatically at her departure, try to cling onto her, become inconsolable, would sometimes throw herself down onto the floor once she's left the room. The friend she left Lauren with told her that Lauren would cry for a long time and could not be comforted. When Ms Coyle returned to pick her up she'd find Lauren to be very quiet, to look at her with a 'frozen' stare and wouldn't reach out for her for comfort. This was quite disturbing to Ms Coyle, who was expecting that Lauren would be pleased to see her, given she was so upset by the separation. What attachment style in Lauren is this vignette illustrating?

Select one:

- Disorganised
- Secure
- Avoidant
- Good enough
- Anxious ambivalent

Your answer is incorrect.

This attachment style describes often unpredictable, confusing and erratic styles of behaving towards the primary attachment figure, often born out of inconsistent or abusive care-giving.

To differentiate the various attachment patterns and to learn more about Ainsworth's experiments, please watch this

The correct answer is: Disorganised

Question 61

Not answered

Marked out of 1.00

Which one among the following is the last stage of Erik Erikson's model of psychosocial development?

Select one:

- Initiative vs guilt
- Identity vs identity confusion
- Industry vs inferiority
- Integrity vs despair
- Intimacy vs isolation

Your answer is incorrect.

Explanation:

Erik Erikson created an original and highly influential theory of psychological development and crisis occurring in periods that extended across the entire life cycle. The last and 8th stage – **integrity vs despair** – takes place from about 60 years until death.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1071-1075.

The correct answer is: Integrity vs despair

Question 62

Not answered

Marked out of 1.00

Which of the following factors protects against child maltreatment in families?

Select one:

- Personality disorder in parents
- History of childhood maltreatment in parents
- Extended family/social networks
- High expectations of parents
- Young age of parents

Your answer is incorrect.

Explanation:

Protective factors for child abuse and neglect include extended family networks and strong, positive social networks. Positive support from various social networks can provide a buffer to reduce caregiver stress and promote caregiver wellbeing, reducing overall maltreatment risk.

Ref: Lamont A, Price-Robertson R. Risk and protective factors for child abuse and neglect. 2013.

Austin AE et al. Risk and protective factors for child maltreatment: A review. Current epidemiology reports. 2020 Dec;7(4):334-42.

The correct answer is: Extended family/social networks

Question 63

Not answered

Marked out of 1.00

According to Margaret Mahler's original theory of separation, what phase of development applies to a newborn child?

Select one:

- Separation-individuation phase
- Rapprochement phase
- Object constancy phase
- Normal autism
- Symbiosis

Your answer is incorrect.

Explanation:

Margaret Mahler proposed a theory to describe how young children acquire a sense of identity separate from that of their mothers'. The first stage is 'normal autism' from birth to 2 months. In this stage, periods of sleep outweigh periods of arousal in a state reminiscent of intrauterine life.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 892-893.

The correct answer is: Normal autism

Question 64

Not answered

Marked out of 1.00

Which of the following statements is true of adolescence?

Select one:

- Fears of rejection for the peer group peaks around the age of 17
- Negative identity is more common in societies that mark transition to adulthood by initiation ceremonies
- Risk taking behaviour is less common in societies that mark transition to adulthood by initiation ceremonies
- The cultural ideal hypothesis suggests that boys move further away from their physical ideal earlier than girls
- Delinquency or offending behaviour reaches a peak in early teens and declines thereafter

Your answer is incorrect.

Initiation is a rite of passage ceremony marking entrance or acceptance into a group or society. It could also be a formal admission to adulthood in a community or one of its formal components. Risk-taking behaviour and negative identity are less common in societies that mark the transition to adulthood by initiation ceremonies. Delinquency or offending behaviour reaches a peak in late teens (17 years) and declines after that.

The correct answer is: Risk taking behaviour is less common in societies that mark transition to adulthood by initiation ceremonies

Question 65

Not answered

Marked out of 1.00

The most common type of attachment behaviour seen in young children is

Select one:

- Anxious resistant
- Secure
- Anxious avoidant
- Absence of attachment
- Disorganised

Your answer is incorrect.

Most children are securely attached (60-70% of children). 15-20% show insecure-avoidant attachment. 10-15% show insecure-ambivalent attachment. 5-10% are disorganised in their attachment style.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 659.

The correct answer is: Secure

Question 66

Not answered

Marked out of 1.00

The game of peek-a-boo becomes characteristically evident during which stage of cognitive development in a child?

Select one:

- Informal operations
- Sensorimotor
- Preoperational
- Formal operations
- Concrete operations

Your answer is incorrect.

Explanation:

In the sensorimotor stage (birth to 2 years), infants learn to differentiate themselves from the world. They also become able to maintain a mental image of an object, even when it is not visible (**object permanence**). This refers to an understanding that an object that disappears from the field of perception has not ceased to exist; if searched for this object can be found, or it will reappear. It is during this stage that children play peek-a-boo.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1007-1008.

The correct answer is: Sensorimotor

Question 67

Not answered

Marked out of 1.00

Mr Richard Parker is a 64-year-old man who has a history of paranoid schizophrenia and has been stable on the same medication regime for several years. He has recently noticed that his skin has a bluish tinge, particularly following a holiday abroad. Which of the following medications is most likely to be the cause?

Select one:

- Promazine
- Haloperidol
- Fluphenazine
- Flupentixol
- Chlorpromazine

Your answer is incorrect.

Explanation:

Chlorpromazine is known to cause abnormal oculocutaneous pigmentation in sun-exposed areas. It can result in abnormal pigmentation of the eyelids, cornea, and lens, as well as blue-grey pigmentation of the skin.

Ref: Huff LS et al. Chlorpromazine-induced skin pigmentation with corneal and lens opacities. Cutis. 2014 May 1;93(5):247-50.

Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 519.

The correct answer is: Chlorpromazine

Question 68

Not answered

Marked out of 1.00

Sandy is currently being treated for psychotic depression. She has a history of sudden cardiac death in her family and has read that some psychiatric medications can increase the risk of prolonged QTc and torsade de pointes. She has been started on an antipsychotic but is discussing with her psychiatrist which antidepressant to start. Which of the following antidepressant is most likely to cause QTC prolongation?

Select one:

- Fluoxetine
- Citalopram
- Sertraline
- Venlafaxine
- Mirtazapine

Your answer is incorrect.

Explanation:

Antidepressant	Effect on QTc (adapted from Maudsley Prescribing Guidelines, 2021. pp. 383-5.)
Agomelatine	Single case of QTc prolongation
Bupropion	QTc shortening, but prolongation has been reported in cases of overdose
Citalopram/escitalopram	Dose-related increase in QTc
Duloxetine	Isolated reports of prolongation
Fluoxetine	No effect on QTc interval
Fluvoxamine	No significant effect on QTc
Levomilnacipran	No effect on QTc interval
Lofepramine	Can possibly prolong QTc interval at higher doses
MAOIs	Unclear but may shorten QTc interval
Milnacipran	No effect on QTc
Mirtazapine	No effect on QTc
Moclobemide	No effect on QTc interval in normal doses; prolongation in overdose
Paroxetine	No effect on QTc interval
Reboxetine	No effect on QTc
Sertraline	No effect on QTc interval at standard doses; small increase at 400mg/day
Trazodone	Can prolong QTc interval
Tricyclics	Prolongation of QTc interval and QRS interval
Venlafaxine/desvenlafaxine	Possible prolongation in overdose, but very rare
Vilazodone	No effect, even in overdose
Vortioxetine	No effect

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 383-385.

The correct answer is: Citalopram

Question 69

Not answered

Marked out of 1.00

Miss Brown is a 27-year-old lady with schizoaffective disorder who has been clinically stable on risperidone for two years. She has recently reported galactorrhoea and been found to have raised prolactin. Which of the following would be the most appropriate strategy for managing her raised prolactin?

Select one:

- Change to paliperidone
- Stop risperidone
- Start estrogen and progestin
- Prescribe bromocriptine
- Consider addition of aripiprazole

Your answer is incorrect.

Explanation:

If treatment of hyperprolactinaemia is required, switching to an antipsychotic with a lower liability for prolactin elevation is usually the first choice although switching always carries a risk of destabilising the illness and relapse. An alternative is to add aripiprazole to existing treatment. Aripiprazole lowers prolactin levels in a dose-dependent manner: 3mg/day is effective but 6mg/day more so. Higher doses appear unnecessary. Other strategies to reduce long-term risk to bone mineral density should also be discussed (e.g. stopping smoking, increasing weight-bearing exercise, and ensuring adequate calcium and vitamin D3 intake).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 169.

The correct answer is: Consider addition of aripiprazole

Question 70

Not answered

Marked out of 1.00

John is a 25-year-old man with treatment resistant schizophrenia. He is due to start an inpatient clozapine titration. He wants to know the rationale for regular blood tests and whether he is at higher risk of problems with his white blood cells. Which of the following is a risk factor for agranulocytosis in patients taking clozapine?

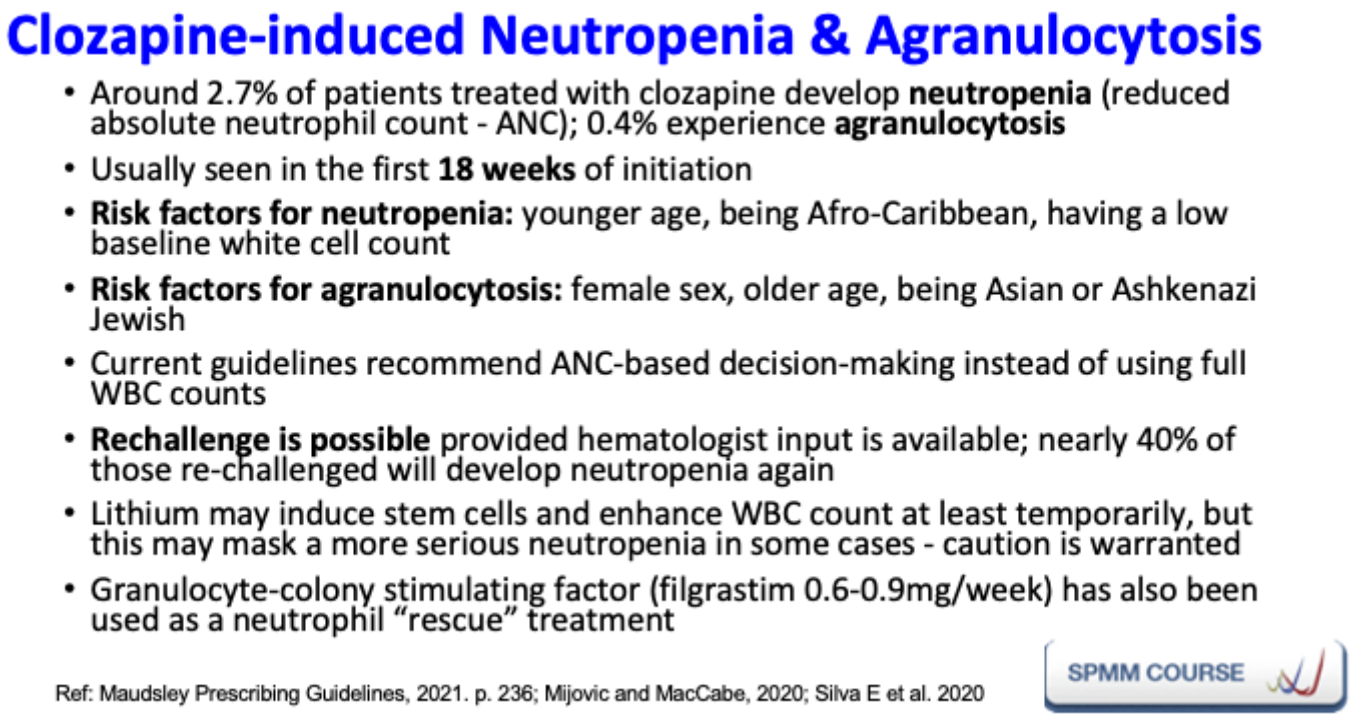
Select one:

- Young age
- Female gender
- More than 1 year of clozapine treatment
- Caucasian ethnicity
- Pre-existing Benign Ethnic Neutropenia

Your answer is incorrect.

Explanation:

Risk factors for agranulocytosis are different from those for neutropenia.



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 236.

Mijovic A, MacCabe JH. Clozapine-induced agranulocytosis. Annals of Hematology. 2020 Aug 20:1-6.

Silva E et al. Clozapine rechallenge and initiation despite neutropenia-a practical, step-by-step guide. BMC psychiatry. 2020 Dec;20:1-9.

The correct answer is: Female gender

Question 71

Not answered

Marked out of 1.00

QT prolongation is a significant side effect of numerous medications as it can lead to torsades de pointes, a fatal arrhythmia. Which of the following medications is associated with QT prolongation?

Select one:

- Ramipril
- Ranitidine
- Digoxin
- Furosemide
- Amlodipine

Your answer is incorrect.

Numerous independent risk factors for QTc interval prolongation and TdP have been identified, and include female sex, hypokalemia, hypomagnesemia, acute myocardial infarction, sepsis, supratherapeutic concentrations of QTc interval-prolonging drugs, rapid intravenous infusion of QTc interval-prolonging drugs, concomitant administration of two or more QTc interval-prolonging drugs, concomitant administration of a loop diuretic, bradycardia, heart failure with reduced ejection fraction, and ion channel polymorphisms. Ref: Tisdale JE. What causes some patients with drug-induced QT interval prolongation to develop torsades de pointes but not others? The elusive missing link. Drugs & aging. 2014 Aug 1;31(8):577-9.

The correct answer is: Furosemide

Question 72

Not answered

Marked out of 1.00

Mr. James is a 69-year-old man with Parkinson's disease on Pramipexole who has recently presented with impulse control problems including pathological gambling and some compulsive buying. His neurologist reduces his pramipexole but very quickly he appears to develop dopamine agonist withdrawal syndrome (DAWS). Which of the following symptoms are not usually seen in DAWS?

Select one:

- Hypertension
- Depression
- Agitation
- Insomnia
- Diaphoresis

Your answer is incorrect.

Explanation:

Reduction and discontinuation of dopamine agonists such as pramipexole may lead to deterioration in motor and non-motor performance. Symptoms of restless leg syndrome may be exacerbated or unmasked for the first time, as can akathisia as well as dopamine responsive non-motor symptoms such as depression, pain, apathy, and sleep disorders. Dopamine agonist withdrawal syndrome (DAWS) may occur in 15-24% of patients exhibiting neuropsychiatric symptoms (anxiety, dysphoria, panic attacks, depression, irritability, and fatigue) and autonomic symptoms (orthostatic hypotension and perspiration).

Ref: Koschel J et al. Implications of dopaminergic medication withdrawal in Parkinson's disease. Journal of Neural Transmission. 2021 Jul 29:1-0.

The correct answer is: Hypertension

Question 73

Not answered

Marked out of 1.00

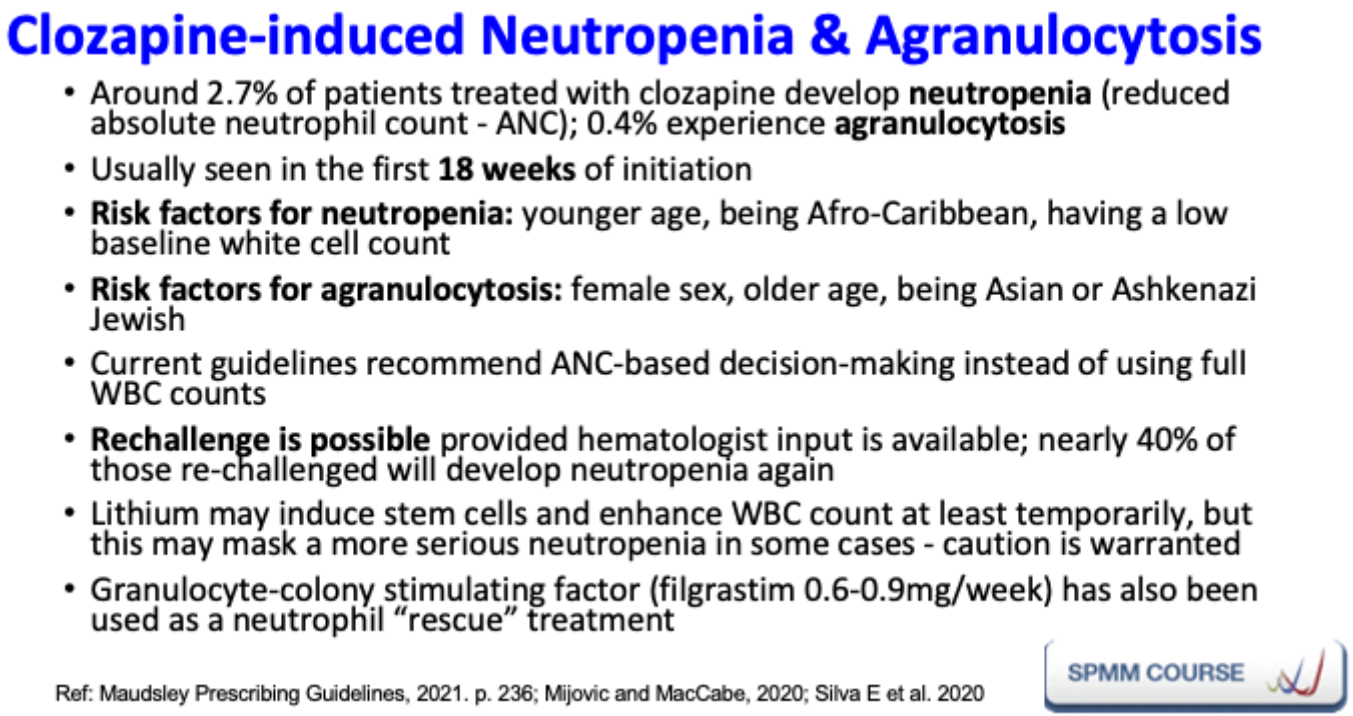
What is a risk factor for developing agranulocytosis with clozapine use?

Select one:

- Female sex
- Younger age
- Caucasian
- Low baseline white cell count
- Long term clozapine use

Your answer is incorrect.

Explanation:



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 236.

Mijovic A, MacCabe JH. Clozapine-induced agranulocytosis. Annals of Hematology. 2020 Aug 20:1-6.

Silva E et al. Clozapine rechallenge and initiation despite neutropenia-a practical, step-by-step guide. BMC psychiatry. 2020 Dec;20:1-9.

The correct answer is: Female sex

Question 74

Not answered

Marked out of 1.00

Ben Staines is a 42-year-old man with a diagnosis of schizophrenia, who has been stable on Clozapine for the past year. He presents to clinic reporting low mood for the past three weeks, with loss of energy, reduced interest in life. You consider starting him on an antidepressant. Which is best to avoid, given that he is on Clozapine?

Select one:

- Fluoxetine
- Selegiline
- Amitriptyline
- Sertraline
- Escitalopram

Your answer is incorrect.

Explanation:

Fluoxetine is a potent inhibitor of CYP2D6 and can raise clozapine levels, resulting in increased seizure risk with clozapine.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 308-309.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021.

The correct answer is: Fluoxetine

Question 75

Not answered

Marked out of 1.00

In relation to weight gain with antipsychotics, which of the following statements is true:

Select one:

- It is more significant in children
- It is insignificant
- Occurs more with first generation antipsychotics
- Occurs less with second generation antipsychotics
- It is more significant in adults

Your answer is incorrect.

Explanation:

An increased susceptibility to antipsychotic-induced weight gain has been repeatedly documented in children, particularly rapid weight gain. Furthermore, antipsychotic medications touted to be weight-neutral in adults can cause substantial weight gain in children.

Ref: Libowitz MR, Nurmi EL. The burden of antipsychotic-induced weight gain and metabolic syndrome in children. Frontiers in Psychiatry. 2021 Mar 12;12:275.

The correct answer is: It is more significant in children

Question 76

Not answered

Marked out of 1.00

Duloxetine is an antidepressant used in the UK. Which of the following is true about it?

Select one:

- Doses of 120mg are routinely used for neuropathic pain
- Patients with gastric bypass surgery may need a lower dose
- It is a dual action antidepressant
- Smoking has no impact on the plasma levels and efficacy
- It has a half-life of around 36 hours

Your answer is incorrect.

Explanation:

- Duloxetine, along with venlafaxine, is sometimes termed as a dual action antidepressant (acts on both serotonin and noradrenaline).
- The elimination half-life of duloxetine is approximately 12 hours.
- Usual dosage ranges are 40-60mg/day in 1-2 doses for depression, 60mg once daily for diabetic peripheral neuropathy and fibromyalgia, and 40mg twice daily for stress urinary incontinence.
- Studies have not demonstrated increased efficacy of duloxetine beyond 60mg/day. Some patients may require up to or more than 120mg/day, but clinical experience is quite limited with high dosing. In neuropathic pain and fibromyalgia, doses above 60mg/day have been associated with increased side effects without an increase in efficacy.

- Plasma levels of duloxetine can be reduced by up to 50% in smokers (Pg 857, Maudsley 14th) and 42% lower after gastric bypass surgery (Pg 823, Maudsley 14th). Higher dose may be necessary for these patients to achieve therapeutic benefits.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 265-266.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 823, 857.

The correct answer is: It is a dual action antidepressant

Question 77

Not answered

Marked out of 1.00

Laura is prescribed a new antidepressant, vortioxetine. Which receptor does vortioxetine exert its action on?

Select one:

- 5HT6
- 5HT3
- 5HT2
- D4
- D2

Your answer is incorrect.

Explanation:

Vortioxetine causes serotonin transporter (SERT) inhibition as well as having antagonist actions at 5HT3 and 5HT7 receptors, with agonist actions at 5HT1A receptors and weak partial agonist to antagonist actions at 5HT1B/D receptors. This unique blend of pharmacological actions leads to downstream release of many different neurotransmitters, and these actions hypothetically lead to antidepressant effects in unipolar depression, characterised by robust pro-cognitive actions, especially improving processing speed.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 315.

The correct answer is: 5HT3

Question 78

Not answered

Marked out of 1.00

Which one among the following is a pharmacodynamic concept?

Select one:

- Excretion
- Distribution
- Metabolism
- Absorption
- Agonism

Your answer is incorrect.

Explanation:

Pharmacokinetics is the study of how the body acts upon drugs, especially to absorb, distribute, metabolise, and excrete them. Pharmacodynamic actions at specific drug targets are known as the mechanism of action of psychotropic drugs, and account for most of the therapeutic effects and side effects of drugs. Agonism, receptor half-life, occupancy, dose-response relationship are pharmacodynamic concepts.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 49.

The correct answer is: Agonism

Question 79

Not answered

Marked out of 1.00

Sasha has returned home from a night out, during which she drank 12 units of alcohol. Her housemate is very worried about her, as she appears barely conscious, and decides to take her to the Emergency Department. You are aware that alcohol is eliminated in a constant amount per unit time, and that this constant does not vary according to plasma concentration of alcohol. What type of elimination kinetics is this describing?

Select one:

- First order
- Linear
- Zero order
- Second order
- Exponential decay

Your answer is incorrect.

In zero order kinetics (also known as non linear kinetics) the elimination rate saturates at high drug concentrations, as occurs with alcohol, phenytoin, and salicylates, among others.

The correct answer is: Zero order

Question 80

Not answered

Marked out of 1.00

The effectiveness of certain psychotropic medications can be affected by pharmacokinetics. Which of the following is an aspect of pharmacokinetics?

Select one:

- Half-life
- Dose-response relationship
- Chelation
- Absorption
- Agonism

Your answer is incorrect.

Explanation:

Pharmacokinetics is the study of how the body acts upon drugs, especially to absorb, distribute, metabolise, and excrete them. Agonism, half-life, dose-response relationship and chelation are pharmacodynamic concepts.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 49.

The correct answer is: Absorption

Question 81

Not answered

Marked out of 1.00

Psychotropic medications will sometimes enter adipose tissue and become concentrated there. What properties of a drug aid its movement into fat tissue?

Select one:

- Methylation
- Hydrophobia
- Oxidation
- Chemical polarity
- Lipophilic

Your answer is incorrect.

Lipophilic compounds enter adipose tissue more easily. The other states described would make crossing a cell membrane more difficult.

The correct answer is: Lipophilic

Question 82

Not answered

Marked out of 1.00

Pharmacokinetics in the elderly is different to those of younger individual due to the physiological changes associated with aging. Which of the following is true regarding the volume of distribution in elderly individuals?

Select one:

- Lower for lipophilic and hydrophilic medications
- Greater for hydrophilic medications
- Greater for lipophilic medications
- Lower for lipophilic medications
- Greater for lipophilic and hydrophilic medications

Your answer is incorrect.

Explanation:

Older adults have more body fat, less body water, and less albumin than younger adults. This leads to an increased volume of distribution and a longer duration of action for some fat-soluble drugs (e.g. diazepam), higher concentrations of drugs at the site of action (e.g. digoxin), and a reduction in the amount of drug bound to albumin (increased amounts of active 'free drug'; e.g. warfarin, phenytoin). There tends to be a greater volume of distribution for lipophilic drugs and lower for hydrophilic medications.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 602.

The correct answer is: Greater for lipophilic medications

Question 83

Not answered

Marked out of 1.00

Pregnancy is a time of significant physiological change and this can have a significant impact upon the effect of medications. Which of the following is true regarding medication metabolism in pregnancy?

Select one:

- Reduced renal blood flow
- Gastric emptying is more rapid
- Increased plasma albumin
- Increased volume of distribution
- Reduced hepatic blood flow

Your answer is incorrect.

The volume of distribution for hydrophilic and lipophilic drugs is increased in pregnancy due to an increase in adipose tissue, blood volume and extracellular fluid.

The correct answer is: Increased volume of distribution

Question 84

Not answered

Marked out of 1.00

Mrs. Jackson is an 89-year-old lady who has been taking temazepam for many years. Which of the following is correct regarding pharmacokinetics in older people?

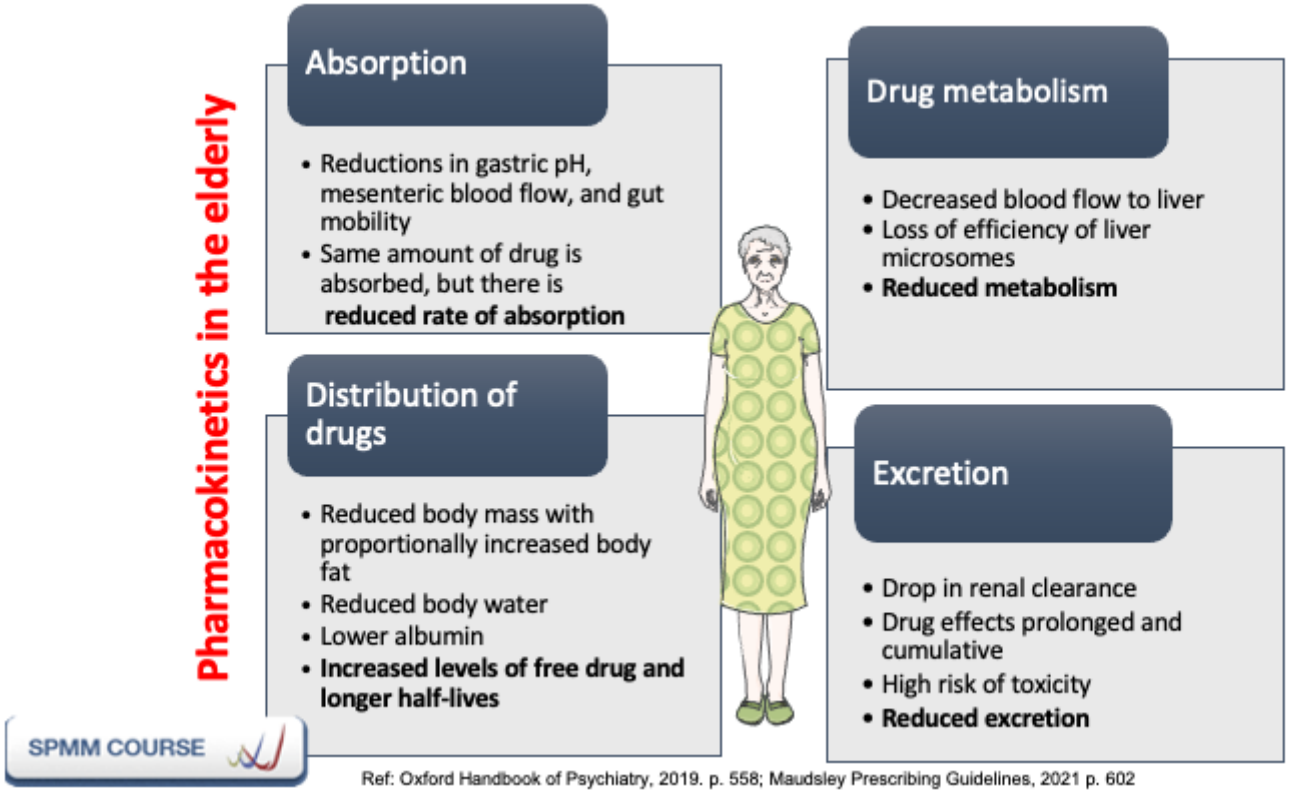
Select one:

- Increase in the amount of drug bound to albumin
- Dosage reduction is usually not required
- Drugs excreted by the kidney rarely accumulate
- Drugs are absorbed from the gut more quickly
- Longer duration of action for fat soluble drugs

Your answer is incorrect.

Explanation:

Older adults have more body fat, less body water, and less albumin than younger adults. This leads to an increased volume of distribution and a longer duration of action for some fat-soluble drugs (e.g. diazepam), higher concentrations of drugs at the site of action (e.g. digoxin), and a reduction in the amount of drug bound to albumin (increased amounts of active 'free drug'; e.g. warfarin, phenytoin).



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 602.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 558.

The correct answer is: Longer duration of action for fat soluble drugs

Question 85

Not answered

Marked out of 1.00

Mr. Jones visits his GP regarding his insomnia after a recent long-haul flight. His GP speaks with him and does not find any evidence of depression or other sleep disturbance. Mr. Jones is keen to try melatonin. Which one of the following statements is correct regarding the endogenous hormone melatonin?

Select one:

- Melatonin levels peak between 11pm and 3am
- It acts on the mu receptor
- Melatonin levels are reduced in darkness
- It is produced by the pituitary
- It is involved in termination of sleep

Your answer is incorrect.

Explanation:

The pineal gland of the epithalamus releases melatonin, perhaps the best-known circadian modulator. Release of melatonin is dependent on environmental inputs, most notably the light cycle; melatonin increases several hours after the onset of the dark phase of the light cycle, and it remains high until the restoration of the light phase the next day (levels peak between 11pm and 3am). In humans, melatonin's effects are the principal mediator for normal sleep onset several hours into the dark phase of the diurnal cycle. There are three types of receptors for melatonin: 1 and 2 (MT1 and MT2), which are both involved in sleep, and 3, which is actually the enzyme NRH-quinine oxidoreductase 2 and not thought to be involved in sleep physiology.

Ref: Smith PC, Mong JA. Neuroendocrine control of sleep. Neuroendocrine Regulation of Behavior. 2019:353-78.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 439.

The correct answer is: Melatonin levels peak between 11pm and 3am

Question 86

Not answered

Marked out of 1.00

Depot antipsychotics are useful in patients who are often non-concordant with medications or who are unwilling to take tablets. Which of the following depot medications has the longest half-life?

Select one:

- Zuclopenthixol decanoate
- Paliperidone palmitate
- Flupenthixol decanoate
- Olanzapine pamoate monohydrate
- Zuclopenthixol acetate

Your answer is incorrect.

Explanation:

Paliperidone is the major active metabolite of risperidone. Paliperidone palmitate is the ester prodrug of paliperidone available as a monthly and a 3-monthly long-acting injection. Kinetics for paliperidone palmitate are determined by particle size: smaller particles (1-month injectable) versus larger particles (3-month injectable). Half-life for the one-month injectable is 25-49 days. Half-life for the 3-month injectable is 84-95 days (deltoid administration) or 118-139 days (gluteal administration).

Antipsychotic long-acting injectable	Plasma half-life (days)	Dose range (mg/week)	Dosing interval (weeks)	Comments
Aripiprazole	30-46d	300-400mg monthly	Monthly	Does not increase prolactin Oral loading required
Flupentixol decanoate	8-17d	50mg every 4 weeks to 400mg a week	2-4	Maximum licensed dose high relative to other LAIs
Fluphenazine decanoate	7-10d	12.5mg every 2 weeks to 100mg every 2 weeks	2-5	High EPS
Haloperidol decanoate	21d	50-300mg every 4 weeks	4	High EPS
Olanzapine pamoate	30d	150mg every 4 weeks to 300mg every 2 weeks	2-4	Risk of post-injection syndrome
Paliperidone palmitate (monthly)	25-49d	50-150mg monthly	Monthly	Loading dose required at treatment initiation
Paliperidone palmitate (3-monthly)	Deltoid: 84-95d Gluteal: 118-139d	175-525mg every 3 months	3 months	
Pipotiazine palmitate	15d	50-200mg every 4 weeks	4	? Lower incidence of EPS (relative to other FGAs)
Risperidone microspheres	4d	25-50mg every 2 weeks	2	Drug release delayed for 2-3 weeks – oral therapy required
Zuclopenthixol decanoate	19d	200mg every 3 weeks to 600mg a week	2-4	? Slightly better efficacy than LAI FGAs

Adapted from Maudsley Prescribing Guidelines, 2021. pp. 69, 72.

SPMM COURSE

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 587-588.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 69, 72, 83.

The correct answer is: Paliperidone palmitate

Question 87

Not answered

Marked out of 1.00

Tricyclic antidepressants are recognised as being potentially hazardous in overdose, particularly when compared to newer antidepressant medications. Which of the following is true regarding tricyclics in overdose?

Select one:

- They are subject to zero order kinetics
- Bradycardia is common
- Gastric emptying is rapid
- They do not bind to serum proteins
- It is associated with alkalosis

Your answer is incorrect.

Tricyclic antidepressants taken in large quantities altered pharmacokinetics. Their absorption may be delayed by inhibition of gastric emptying and peristalsis. A significant enterohepatic recirculation delays final elimination of a large fraction of the drug. The enzymes responsible for TCA benzyl-hydroxylation can become saturated and thus reduce TCA elimination to zero-order kinetics. They are highly protein-bound although there are more free compounds in overdose. Overdose is associated with acidosis, further delaying clearance, and tachycardia is the most common arrhythmia seen. Ref: Jarvis MR. Clinical pharmacokinetics of tricyclic antidepressant overdose. Psychopharmacology bulletin. 1991;27(4):541-50.

The correct answer is: They are subject to zero order kinetics

Question 88

Not answered

Marked out of 1.00

John Wilson is a 37-year-old Caucasian male with treatment-resistant schizophrenia. He meets with his psychiatrist to discuss the potential side effects of the antipsychotic clozapine. What is the risk of agranulocytosis in a patient taking clozapine after one year?

Select one:

- 5%
- 0.5%
- 0.1%
- 1%
- 0.01%

Your answer is incorrect.

Explanation:

The prevalence of agranulocytosis during clozapine treatment is 0.4%, lower than previously thought. Clozapine-induced agranulocytosis typically occurs within the first 18 weeks of treatment, with few cases occurring beyond 6 months. After 1 year of treatment, residual risk is estimated at 0.39/1000 patient/years (risk of <0.01% after one year).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 236.

Mijovic A, MacCabe JH. Clozapine-induced agranulocytosis. Annals of Hematology. 2020 Aug 20:1-6.

The correct answer is: 0.01%

Question 89

Not answered

Marked out of 1.00

Mr. Davies is a 32-year-old man with a diagnosis of bipolar affective disorder who has required prolonged admissions in the past for episodes of mania. He has now presented to clinic with a severe depressive episode despite being compliant with mood stabilising medication. He is wondering about taking an antidepressant but is worried about the risk of a manic episode and refuses to take olanzapine in combination due to severe weight gain in the past. Which of the following antidepressants is least likely to induce a manic switch while already taking long-term treatment for bipolar disorder?

Select one:

- Sertraline
- Paroxetine
- Vortioxetine
- Phenelzine
- Venlafaxine

Your answer is incorrect.

As reflected in the most recent bipolar guidelines, there are several options, including the addition of olanzapine+fluoxetine, quetiapine, olanzapine, or lamotrigine. But this patient refuses olanzapine.

The meta-analytical literature (Gijsman et al, 2004) provides a rate of switching for tricyclic antidepressants as 10% compared with 3.2% for all other antidepressants combined. MAOIs are also categorised to have a higher switch rate (Yatham, Kusumakar & Kutcher, 2002; Page 24). There is a 4-fold higher switch rate for venlafaxine (29%) compared with SSRIs (sertraline 9%) and bupropion (10%) when used concurrently with a mood stabilizer. Fluvoxamine may have much lower risk and newer agents such as vortioxetine may have a higher switch rate, but further studies are needed to reduce the uncertainty.

Ref: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4321017/ .

Also see Jo YT et al. Diagnostic conversion from unipolar to bipolar affective disorder: a population-based study. Journal of Affective Disorders. 2022 Jan 20.

National Institute for Health and Clinical Excellence. The NICE guideline on the assessment and management of bipolar disorder in adults, children and young people in primary and secondary care. Updated edition 2020.

The correct answer is: Sertraline

Question 90

Not answered

Marked out of 1.00

Sadira is a 36-year-old female from South India. She is an inpatient on your ward and was diagnosed with treatment-refractory schizophrenia five months ago. Since then she has been treated with psychotropic medication, which she initially tolerated well. She says that her auditory hallucinations have lessened in their intensity. Previously a heavy smoker, Sadira managed to give this up recently. During the ward round she complains of feeling increasingly drowsy. You look at her medications to see if she is taking anything that may be contributing to this. Which of the following medications are most likely contributing to her current presentation?

Select one:

- Paracetamol
- Zopiclone
- Sertraline
- Clozapine
- Mirtazapine

Your answer is incorrect.

Explanation:

Tobacco smoke contains polycyclic aromatic hydrocarbons that induce (increase the activity of) certain hepatic enzymes (CYP1A2 in particular). The extent of enzyme induction is determined by the number and type of cigarettes smoked and the degree of smoke inhalation. For some drugs used in psychiatry, smoking significantly reduces drug plasma levels and higher doses are required than in non-smokers. Smoking reduces plasma clozapine levels by up to 50%. When people stop smoking, enzyme activity halves roughly every 2 days. Plasma levels of affected drugs will then rise, sometimes substantially. Dose reduction will usually be necessary.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 856.

The correct answer is: Clozapine

Question 91

Not answered

Marked out of 1.00

Mr. Collins is a 48-year-old man with a diagnosis of alcohol dependence. He previously had 10 years of abstinence after treatment with disulfiram but relapsed 3 years ago. He is keen to have disulfiram again. Which of the following is a side-effect of disulfiram in the absence of alcohol consumption?

Select one:

- Excessive sweating
- Flushing
- Palpitations
- Nausea
- Headache

Your answer is incorrect.

Explanation:

In the absence of alcohol consumption, side effects of disulfiram include halitosis, metallic taste, headache, dermatitis, and sedation.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 598.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 239.

The correct answer is: Headache

Question 92

Not answered

Marked out of 1.00

Mr Edwards is a 60-year-old man who has erectile dysfunction. He attends his GP asking about sildenafil. Which of the following is a common side-effect of this medication?

Select one:

- Constipation
- Diarrhoea
- Nasal congestion
- Polydipsia
- Nausea

Your answer is incorrect.

Explanation:

Common side effects of phosphodiesterase inhibitors (such as sildenafil and tadalafil) are mild headaches, dizziness, and nasal congestion.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 176.

The correct answer is: Nasal congestion

Question 93

Not answered

Marked out of 1.00

A hypertensive patient develops a small focal lesion of one inferior olivary nucleus. Which of the following clinical signs can be expected?

Select one:

- Right Left Disorientation
- Visual Inattention
- Anosognosia
- Oculomotor Apraxia
- Finger Nose Ataxia

Your answer is incorrect.

Inferior Olivary Nucleus has a role in motor coordination by providing CLIMBING FIBERS to the cerebellum.Lesion results in motor incoordination of the contralateral arm and leg (unlike other cerebellar lesions which cause ipsilateral dysfunction)

The correct answer is: Finger Nose Ataxia

Question 94

Not answered

Marked out of 1.00

A 50-year-old man on a neurology ward has non-dominant parietal lobe dysfunction. Identify the test that he is least likely to be able to do?

Select one:

- Reading a sentence
- Serial subtraction of 7s from 100
- Copying intersecting pentagons
- Writing a sentence
- Correctly stating current location

Your answer is incorrect.

The cognitive process tested by asking the patient to copy intersecting pentagons is constructional praxis, which is a feature of non-dominant parietal lobe lesions. The effects of lesion involving non-dominant parietal lobe include visuospatial dysfunction, topographical memory loss, and constructional / dressing apraxia. These disorders could be seen in lesions of either hemisphere but observed more frequently and are of greater severity with lesions of the non-dominant parietal lobe.

The correct answer is: Copying intersecting pentagons

Question 95

Not answered

Marked out of 1.00

Mr. Smith was seen by a clinical psychologist at a memory clinic and was noted to have features of executive dysfunction. Executive function is tested using which of the following tests?

Select one:

- National Adult Reading Test
- Raven's Progressive Matrices
- Wisconsin Card Sorting Test
- Vocabulary Test
- Rey-Osterrieth Test

Your answer is incorrect.

Explanation:

The Wisconsin card sorting task is a test of executive functioning. It is a measure of problem-solving efficiency that is particularly sensitive to executive deficits of perseveration and impaired ability to flexibly generate alternative strategies in response to feedback. The patient has to determine the rule for card allocation and allocate cards accordingly. When the rule changes, a patient with frontal lobe dysfunction is likely to make more errors (tests response inhibition and set shifting).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 53.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 88.

The correct answer is: Wisconsin Card Sorting Test

Question 96

Not answered

Marked out of 1.00

Non-fluent expressive dysphasia is seen in which of the following conditions

Select one:

- Frontotemporal dementia
- Lewy body dementia
- Progressive supranuclear palsy
- Parkinson's disease
- Creutzfeldt Jacob disease

Your answer is incorrect.

Explanation:

Broca aphasia is also called non-fluent expressive dysphasia. There is reduced frequency and fluency of speech and presents with few disjointed words. Comprehension is relatively preserved. In frontotemporal dementia, particularly the progressive non-fluent aphasia subtype, atrophy in Broca area causes non-fluent effortful speech with agrammatism.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 160.

The correct answer is: Frontotemporal dementia

Question 97

Not answered

Marked out of 1.00

Digit span test is administered to patient with suspected cognitive impairment. This test detects problem in which of the following cognitive faculties?

Select one:

- Semantic memory
- Procedural memory
- Premorbid IQ
- Working memory
- Calculation ability

Your answer is incorrect.

Explanation:

Digit span is a test of working memory (immediate) and attention. Immediate (or working) memory refers to the amount of information that can be held in mind (like a phone number or digit span) to be available for immediate use. Digit span requires patients to repeat longer strings of digits increasingly as a way of assessing the ability to process relatively simple information.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 54.

The correct answer is: Working memory

Question 98

Not answered

Marked out of 1.00

The risk of schizophrenia in a monozygotic twin of schizophrenia patient compared to the general population is

Select one:

- 7 times higher
- 2 times higher
- 50 times higher
- 10 times higher
- 3 times higher

Your answer is incorrect.

Explanation:

Genetic factors account for most of the liability to schizophrenia. Heritability estimates range from 60-80%. The risk of developing schizophrenia if one is no relatives affected is 0.5-1%. The risk of developing schizophrenia when one has an affected identical twin is approximately 46%, about 50 times higher than the general population.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 190-191.

The correct answer is: 50 times higher

Question 99

Not answered

Marked out of 1.00

You are seeing a patient in a gender identity clinic. The patient would like to obtain membership of a gender that they experience, which is different from the assigned gender based on sex at birth. They are also keen to be treated as a member of the experienced gender. The most probable description when using ICD-11 nomenclature is

Select one:

- Body dysmorphic disorder
- Transexualism
- Gender incongruence
- Frotteurism
- Fetishistic transvestism

Your answer is incorrect.

ICD-11 Gender Incongruence, in contrast to DSM-5 Gender Dysphoria, is explicitly not a mental disorder as it requires neither distress nor dysfunction as a feature. Notably, Gender Incongruence in Childhood and in Adolescence and Adulthood does not have the label “disorder” anymore, and are moved out of the chapter on Mental, Behavioral or Neurodevelopmental Disorders to a new ICD-11 chapter on Conditions Related to Sexual Health. Further, unlike DSM Gender Dysphoria, a 6-months duration is not required; instead 'several months' is described as a feature.

Despite these differences, the commonalities between ICD-11 and DSM-5 are: (1) a strong dislike or discomfort with the one's sex characteristics, (2) a strong desire to be rid of some of all of one's sex characteristics, (3) a strong desire to have the sex characteristics of the experienced gender, and (4) a strong desire to be treated (to live and be accepted) as a person of the experienced gender.

Ref: Reed, G.M., et al, 2016. Disorders related to sexuality and gender identity in the ICD-11: revising the ICD-10 classification based on current scientific evidence, best clinical practices, and human rights considerations. *World Psychiatry*, *15*(3), pp.205-221.

Robles, R., et al, 2022. Validity of Categories Related to Gender Identity in ICD-11 and DSM-5 Among Transgender Individuals who Seek Gender-Affirming Medical Procedures. *International Journal of Clinical and Health Psychology*, *22*(1), p.100281.

The correct answer is: Gender incongruence

Question 100

Not answered

Marked out of 1.00

The presence of a single apolipoprotein-e4 allele increases the chance of developing Alzheimer's disease by around

Select one:

- Twenty fold
- Two fold
- Fifty fold
- Ten fold
- Three fold

Your answer is incorrect.

A polymorphism in the **Apolipoprotein ε4** (Chromosome 19) is a major risk factor for late- onset Alzheimer disease. Specifically, the presence of ε4 alleles increases risk of **late-onset** AD; people with **one copy** of the gene (ε3/4 heterozygotes) have **3x the incidence** of Alzheimer's disease than no ε4 gene, and people with **two ε4 genes** have the disease **8 times more frequently** than do those with no ε4 gene (some quote a higher risk); **genetic testing** for presence of ε4 is **not currently recommended** as it is also present in people without dementia, and not found in all cases of dementia. This gene promotes the deposition of beta-amyloid protein in the brain, contributing to the formation of senile plaques.

In most populations, the ε3 allele is the most common. However, in familial late-onset Alzheimer disease, the incidence of ε4 is approximately 50%, compared with 16% in normal controls. The ancestral allele ε2, in contrast, is underrepresented.

Epidemiological studies suggest that 30-60% of late-onset Alzheimer disease cases have at least one ε4 allele. The ε4 genotype appears to be a more important risk factor for Alzheimer disease in populations of European and Asian origin when compared with populations of African origin.

Ref: Fernandez CG, Hamby ME, McReynolds ML, Ray WJ. The role of APOE4 in disrupting the homeostatic functions of astrocytes and microglia in aging and Alzheimer's disease. Frontiers in aging neuroscience. 2019 Feb 11;11:14.

The correct answer is: Three fold

Question 101

Not answered

Marked out of 1.00

Culturing cells in conditions of thymidine deprivation has aided in the understanding of which of the following disorders?

Select one:

- Fragile X syndrome
- Asperger disease
- Cri-du-chat syndrome
- Autism
- Down syndrome

Your answer is incorrect.

Explanation:

Fragile X syndrome is associated with a rare fragile and unsteady site on Xq27. This site, named FRAXA, was originally observed as a non-staining gap, break, or constriction in the metaphase chromosomes placed under selective cultural conditions, such as folic acid or thymidine deprivation.

Ciaccio C et al. Fragile X syndrome: a review of clinical and molecular diagnoses. Italian journal of pediatrics. 2017 Dec;43(1):1-12.

The correct answer is: Fragile X syndrome

Question 102

Not answered

Marked out of 1.00

Social anxiety and gaze avoidance are features of which of the following trinucleotide expansion syndrome?

Select one:

- Friedreich Ataxia
- Leber Optic Neuropathy
- Myotonic dystrophy
- Fragile X syndrome
- Huntington disease

Your answer is incorrect.

Explanation:

Fragile X syndrome is associated with a **trinucleotide (CGG) triplet repeat** on the distal part of the long arm of the X chromosome (Xq27.3-FMR1) that leads to the characteristic 'fragile' appearance of the chromosome. FMR1 gene is required for normal neural development; penetrance is low, but greater in males than females (due to the protective effects of the second normal X chromosome in females). **Behavioural features** are similar to those seen in ADHD and autism and include hand flapping/waving, repetitive mannerisms, **shyness**, **gaze avoidance**, poor peer relationships, communication difficulties (e.g. delayed language development, conversational rigidity, perseveration, echolalia, palilalia(repetition of syllables, words, phrases), cluttering, and over-detailed speech), and psychiatric problems (e.g. **prominent depression/anxiety**).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 814.

The correct answer is: Fragile X syndrome

Question 103

Not answered

Marked out of 1.00

One of the following disorders do not involve genetic deletion. Choose the correct answer.

Select one:

- William syndrome
- Angelman syndrome
- Cri-du-chat syndrome
- DiGeorge syndrome
- Friederich ataxia

Your answer is incorrect.

Explanation:

Friedreich ataxia is an autosomal recessive spinocerebellar ataxia. It is the most common inherited ataxia in Europe. In the majority of cases the disease is caused by a homozygous GAA triplet repeat expansion in the frataxin (FXN) gene, and the shorter repeat expansion length correlates with age at onset and disease severity.

Ref: Cook A, Giunti P. Friedreich's ataxia: clinical features, pathogenesis and management. British medical bulletin. 2017 Dec 1:19-30.

The correct answer is: Friederich ataxia

Question 104

Not answered

Marked out of 1.00

Which one of the following is a primary proponent of endophenotypes concept?

Select one:

- Gottesmann
- Freud
- Kraepelin
- Fromm-Reichmann
- Kendler

Your answer is incorrect.

Explanation:

Having broken new ground in the field of polygenicity and epigenetics, Irving Gottesman understood that unraveling the complex genomic basis of the fuzzy and qualitative diagnosis of schizophrenia would be a daunting task. How to make the task simpler, more parsimonious, and more neurobiologically meaningful? To do this, Gottesman introduced the concept of endophenotypes into the schizophrenia literature. Endophenotypes are laboratory-based measures that show deficits in schizophrenia patients and lesser deficits in their first-degree 'unaffected' relatives and are viewed as being more proximal to genes and having a simpler genetic architecture than are 'fuzzy' qualitative diagnostic disorders.

Ref: Braff DL, Tamminga CA. Endophenotypes, epigenetics, polygenicity and more: Irv Gottesman's dynamic legacy. Schizophrenia bulletin. 2017 Jan 1;43(1):10-6.

The correct answer is: Gottesmann

Question 105

Not answered

Marked out of 1.00

The process of meiosis produces

Select one:

- Diploid somatic cells
- Haploid germ cells
- Tetraploid cells
- Triploid cells
- Aneuploid cells

Your answer is incorrect.

Explanation:

Meiosis is a form of cell division in which the nucleus divides into four daughter nuclei, each containing half the chromosomes of the parent nucleus. It consists of two consecutive divisions: during the first, homologous chromosomes pair up, exchange genetic material by crossing over ('recombination'), then during the anaphase part of the cycle move apart by disjunction to form separate haploid nuclei. During the second phase, these daughter nuclei divide by mitosis (with a further anaphase), producing four haploid cells.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Haploid germ cells

Question 106

Not answered

Marked out of 1.00

Which of the following is best termed as an intermediate phenotype with high occurrence in relatives but cannot be justifiably classified as an endophenotype for schizophrenia?

Select one:

- Impaired attentional performance
- Working memory deficits
- Eye movement dysfunctions
- Impaired p50 suppression
- Neurological soft signs

Your answer is incorrect.

Explanation:

Predominantly, neurological soft signs (NSS) have been looked at as endophenotypes (i.e., hardcopies of the genetic liability) and thus a trait phenomenon. However, the literature on NSS in schizophrenia presents at least partially diverging results. On the one hand, research speaks in favour of NSS as being a stable entity (i.e., a trait): NSS are more pronounced in patients compared to their relatives; they are present at disease onset; there is a relationship with the most stable symptoms, namely negative and cognitive ones. On the other hand, there are arguments supporting a fluctuating aspect, especially of motor signs. Only the presence of a trait can be used as an endophenotype or a predictor, but not the quantitative expression (i.e., the state). Thus, the usage of NSS as an endophenotype has to be called into question.

The most established endophenotypes for schizophrenia are changes in neurocognition (attention deficits, working and verbal declarative memory dysfunctions) and neurophysiology (pre-pulse inhibition, anti-saccade impairment, event-related potential deficits) (Donati et al., 2020).

Ref: Bachmann S, Schröder J. Neurological soft signs in schizophrenia: an update on the state-versus trait-perspective. Frontiers in psychiatry. 2018 Jan 8;8:272.

Donati FL, D'Agostino A, Ferrarelli F. Neurocognitive and neurophysiological endophenotypes in schizophrenia: An overview. Biomarkers in Neuropsychiatry. 2020 Dec 1;3:100017.

The correct answer is: Neurological soft signs

Question 107

Not answered

Marked out of 1.00

The Kluver-Bucy syndrome is typically associated with lesions of which of the following structures?

Select one:

- Amygdala
- Medial dorsal thalamus
- Lateral hypothalamus
- Globus pallidus
- Subthalamic nucleus

Your answer is incorrect.

Explanation:

In 1937, Heinrich Kluver and Paul Bucy described a dramatic behavioural syndrome in monkeys after bilateral temporal lobectomy. The full Kluver-Bucy syndrome (KBS) – hyperorality, placidity, hypermetamorphosis, dietary changes, altered sexual behaviour, and visual agnosia – is evident within 3 weeks following operation. By the 1970s, it was clear that the major symptoms of KBS are produced by destroying either the temporal neocortex or the amygdala bilaterally. The clinical features of KBS in humans are similar to those in monkeys, but the full syndrome is rarely seen, probably because the anterior temporal lobe dysfunction is usually less severe than that following total temporal lobe ablation in monkeys.

Ref: Lanska DJ. The Klüver-Bucy syndrome. Neurologic-Psychiatric Syndromes in Focus-Part I. 2018;41:77-89.

The correct answer is: Amygdala

Question 108

Not answered

Marked out of 1.00

The last part of the brain to mature is

Select one:

- Posterior parietal pole
- Occipital pole
- Anterior parietal pole
- Anterior frontal pole
- Hippocampus

Your answer is incorrect.

The last area of the brain to develop is the frontal lobe. The last region of the brain to mature is pre-frontal cortex (anterior frontal pole). While it may be fully formed in size, various parts within the structure of the brain continue to develop well into a person's twenties. The adolescent brain is immature and in a dynamic state of development. Frontal brain development is also a factor related to impulsive behavior among teens.

The correct answer is: Anterior frontal pole

Question 109

Not answered

Marked out of 1.00

Hypothetical use of cholecystokinin has been studied in which of the following disorders?

Select one:

- Bulimia
- Conduct disorder
- Childhood schizophrenia
- Autism
- ADHD

Your answer is incorrect.

Explanation:

Neurobiological disturbances are present in individuals with bulimia nervosa and may increase the probability of binge eating. In particular, individuals with the disorder tend to display delayed gastric emptying, increased stomach capacity, and reduced secretion of cholecystokinin, a peptide hormone released by the small intestine that helps to signal satiety during food consumption.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 479.

The correct answer is: Bulimia

Question 110

Not answered

Marked out of 1.00

The enzyme involved in degradation of noradrenaline in presynaptic neurons is

Select one:

- GABA transaminase
- Monoamine oxidase
- Homovanillic acid
- Vanillyl mandelic acid
- 3-methoxy, 4-hydroxy, phenylglycol

Your answer is incorrect.

Explanation:

Norepinephrine (NE) action is terminated by two principal destructive or catabolic enzymes that turn NE into inactive metabolites. The first is monoamine oxidase A or B, which is located in mitochondria in the presynaptic neuron and elsewhere. The second is catechol-O-methyltransferase (COMT), which is thought to be located largely outside of the presynaptic nerve terminal. The action of NE can be terminated not only by enzymes that destroy NE, but also by a transport pump for NE that removes it from acting in the synapse without destroying it. That transport pump is sometimes called the 'NE transporter' or 'NET' and sometimes the 'NE reuptake pump'.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 253-254.

The correct answer is: Monoamine oxidase

Question 111

Not answered

Marked out of 1.00

Which of the following receptors requires the simultaneous binding of two different agonists for activation?

Select one:

- GABA B
- Muscarinic M3
- GABA A
- Mu Opiate
- NMDA

Your answer is incorrect.

Explanation:

Glutamate systems are curious in that one of the key receptors for glutamate (the NMDA receptor) requires a co-transmitter in addition to glutamate in order to function. The co-transmitter is either the amino acid glycine or another amino acid, closely related to glycine, known as D-serine.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 97-98.

The correct answer is: NMDA

Question 112

Not answered

Marked out of 1.00

Which of the following pathological bodies is made up of ubiquitin and alpha synuclein and is associated with loss of dopaminergic neurons in basal ganglia?

Select one:

- Hirano bodies
- Lewy bodies
- Neurofibrillary tangles
- Barr bodies
- Prion bodies

Your answer is incorrect.

Explanation:

The key constituent of Lewy bodies is alpha-synuclein. When aggregates of misfolded, overexpressed alpha-synuclein accumulate in neurons or surrounding glial cells, Lewy bodies are formed. Lewy bodies are immunoreactive for alpha-synuclein as well as for ubiquitin and alpha-beta-crystallin. Neuropathic changes of dementia with Lewy bodies include the presence of Lewy bodies and Lewy neurites in the cerebral cortex as well as in the brainstem. These are often accompanied by some degree of Alzheimer disease neuropathologic changes, often with a moderate beta-amyloid plaque burden but relatively lower tangle burden.

Ref: Sanford AM. Lewy body dementia. Clinics in geriatric medicine. 2018 Nov 1;34(4):603-15.

Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. pp. 6-7, 197.

The correct answer is: Lewy bodies

Question 113

Not answered

Marked out of 1.00

A 55 year old lady presents with weakness of her legs and hands. She feels clumsy with a history of falls. On examination she has fasciculations in her tongue and memory deficits. She is diagnosed to have amyotrophic lateral sclerosis. The primary pathological change will be noted in

Select one:

- Motor neurons
- Neuromuscular junction
- Myocytes
- Spinal autonomic ganglion
- Sensory neurons

Your answer is incorrect.

Explanation:

Amyotrophic lateral sclerosis (ALS) is characterised by degeneration of both upper motor neurons (UMNs) and lower motor neurons (LMNs). While the balance between the severity and pace of involvement of these two sets of neurons may vary from case to case, the combination is characteristic of ALS. In many cases, the disease begins with involvement of spinal LMNs, manifesting as weakness in the arms or legs. In others, ALS begins with bulbar symptoms and may be termed progressive bulbar palsy.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. p. 215.

The correct answer is: Motor neurons

Question 114

Not answered

Marked out of 1.00

Which of the following accounts for spongiform appearance in CJD?

Select one:

- Accumulation of prion proteins
- Neuropil vacuolation
- Trapped air in parenchymal cells
- Neuronal loss
- Neuropil loss

Your answer is incorrect.

Explanation:

Neuropathologic findings in Creutzfeldt-Jakob disease (CJD) are characterised by spongiform changes secondary to neuropil vacuolation. Vacuolated neurons and/or vacuolated neuropil is observed typically in CJD.

Ref: Iwasaki Y. Creutzfeldt-Jakob disease. Neuropathology. 2017 Apr;37(2):174-88.

Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. p. 4.

The correct answer is: Neuropil vacuolation

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